FACULTY OF COMPUTING, ELECTRONICS AND SCIENCE ASSESSMENT

NETWORK TECHNOLOGIES

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Introduction

This report aims to provide a comprehensive comparison of the IPv6 Routing Information Protocol (RIPng) and the Open Shortest Path First version 3 (OSPFv3) in Single Area mode. The evaluation will include the format and usage of IPv6 addresses, configuration commands for router interfaces, and the detailed implementation of both RIPng and OSPFv3 protocols. The assessment will be supported by Packet Tracer simulations.

1. Format and Functionality of IPv6 Addresses:

IPv6 addresses, a crucial component of modern networking, consist of 128 bits, a significant expansion compared to the limited IPv4 addressing space. This evolution aims to address the imminent scarcity of IPv4 addresses, especially with the surge in interconnected devices due to the Internet of Things (IoT).

IPv6 Address Structure

An IPv6 address is a 128-bit numerical string organized into eight groups of 16 bits each, separated by colons. This structure distinguishes it from its predecessor, IPv4. The address comprises two main components: the network (first 64 bits) and the node (subsequent 64 bits). The network portion aids in routing, while the node portion identifies the interface. Further subdivision into 48-bit and 16-bit segments allows for IP address allocation and internal subnet management.

IPv6 Address Representation

Representing an IPv6 address involves a sequence of four hexadecimal digits in each 16-bit group, separated by colons. To reduce address length, leading zeros and consecutive sequences of zeros can be excluded. Three key components of an IPv6 address are the routing prefix, subnet ID, and interface ID.

Types of IPv6 Addresses

IPv6 addresses come in various types, each serving distinct purposes:

1. **Global Unicast Addresses:** Addresses with the prefix "2001:" are routable, akin to public IPv4 addresses.
2. **Unicast Addresses:** These identify individual nodes on the network.
3. **Anycast Identifiers:** Used in Anycast networking to assign identifiers to distributed network interfaces.
4. **Multicast Addresses:** Facilitate the simultaneous transmission of a packet to multiple recipients.
5. **Link Local Addresses (fe80:):** Restricted to private networks, serving communication within a specific locality.
6. **Unique Local Addresses:** Internal addresses not routable on the internet, serving devices unconnected to the web.

In summary, IPv6 addresses, with their expanded bit space, offer a solution to the address depletion challenge faced by IPv4. The structured format and diverse address types cater to the intricate needs of modern networking.

Decoding IPv6 Address Notation and Purpose

IPv6 addresses play a pivotal role in networking, providing a vast pool of 128-bit addresses compared to the limitations of IPv4. This evolution responds to the increasing demand for addresses driven by the Internet of Things (IoT) and the impending exhaustion of IPv4 addresses.

Anatomy of IPv6 Addresses

IPv6 addresses adopt a unique structure, comprising eight groups of 16 bits each, totaling 128 bits. These are divided into the network (first 64 bits) and node (subsequent 64 bits) components. Further segmentation into 48-bit and 16-bit segments facilitates IP address allocation and internal subnet management by network administrators.

Representing IPv6 Addresses

Representation involves grouping hexadecimal digits into four-digit sequences within each 16-bit block, separated by colons. The exclusion of leading zeros and consecutive zero sequences streamlines address presentation. Key components include the routing prefix, subnet ID, and interface ID, contributing to address functionality.

Categories of IPv6 Addresses

IPv6 addresses serve various purposes:

1. **Global Unicast Addresses (2001:):** Routable addresses comparable to public IPv4 addresses.
2. **Unicast Addresses:** Identifiers for individual nodes on the network.
3. **Anycast Identifiers:** Facilitate Anycast networking by assigning identifiers to distributed interfaces.

4. **Multicast Addresses:** Enable simultaneous packet transmission to multiple recipients.
5. **Link Local Addresses (fe80:):** Restricted to local networks, promoting communication within a specific area.
6. **Unique Local Addresses:** Internal addresses for non-internet-connected devices.

In essence, IPv6 addresses, with their expanded bit space, resolve IPv4 address scarcity issues. Their organized format and diverse types accommodate the complex needs of modern networking.

2. IPv6 RIP (RIPng) Implementation

Topology Overview

The final Packet Tracer topology for IPv6 implementation involves routers and PCs interconnected in a network. To ensure seamless communication, IPv6 addresses need to be configured, and the Routing Information Protocol for IPv6 (RIPng) is employed for routing.

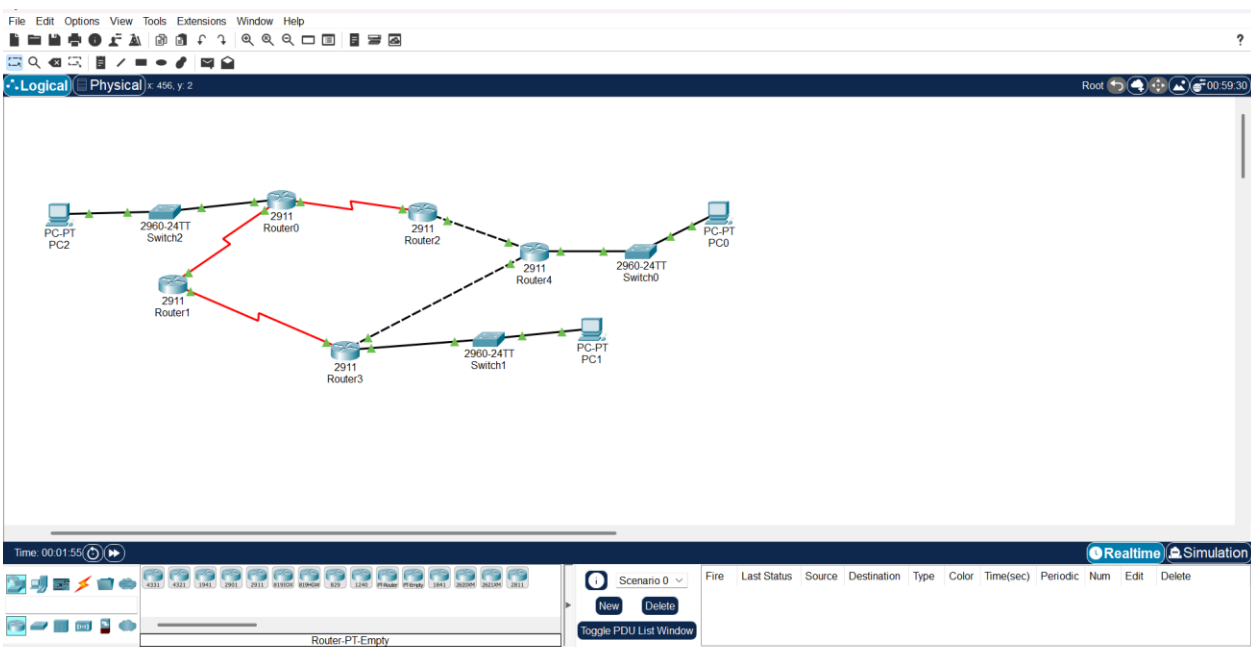


Figure 1: cisco Workspace

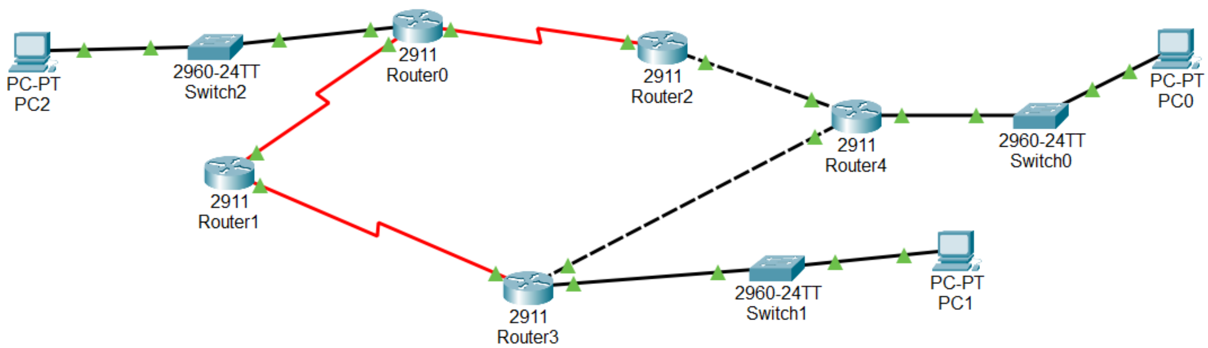


Figure 2: Final Network Topology

Topology Configuration Steps

1. Configuring IPv6 Addresses on Router Interfaces

- Begin by configuring IPv6 addresses on each router interface. Use the "ipv6 enable" command after globally activating IPv6.

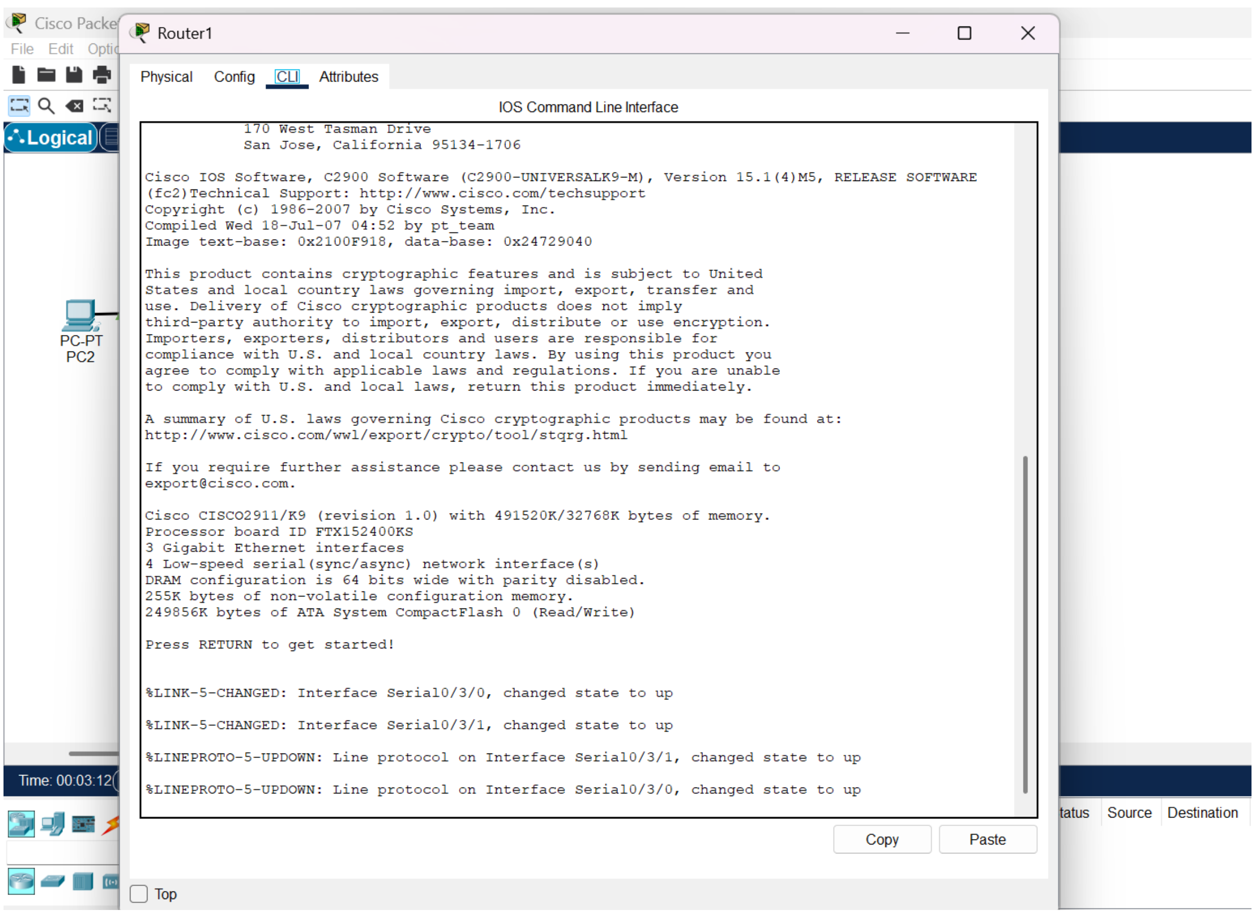


Figure 3: Router1 CLI

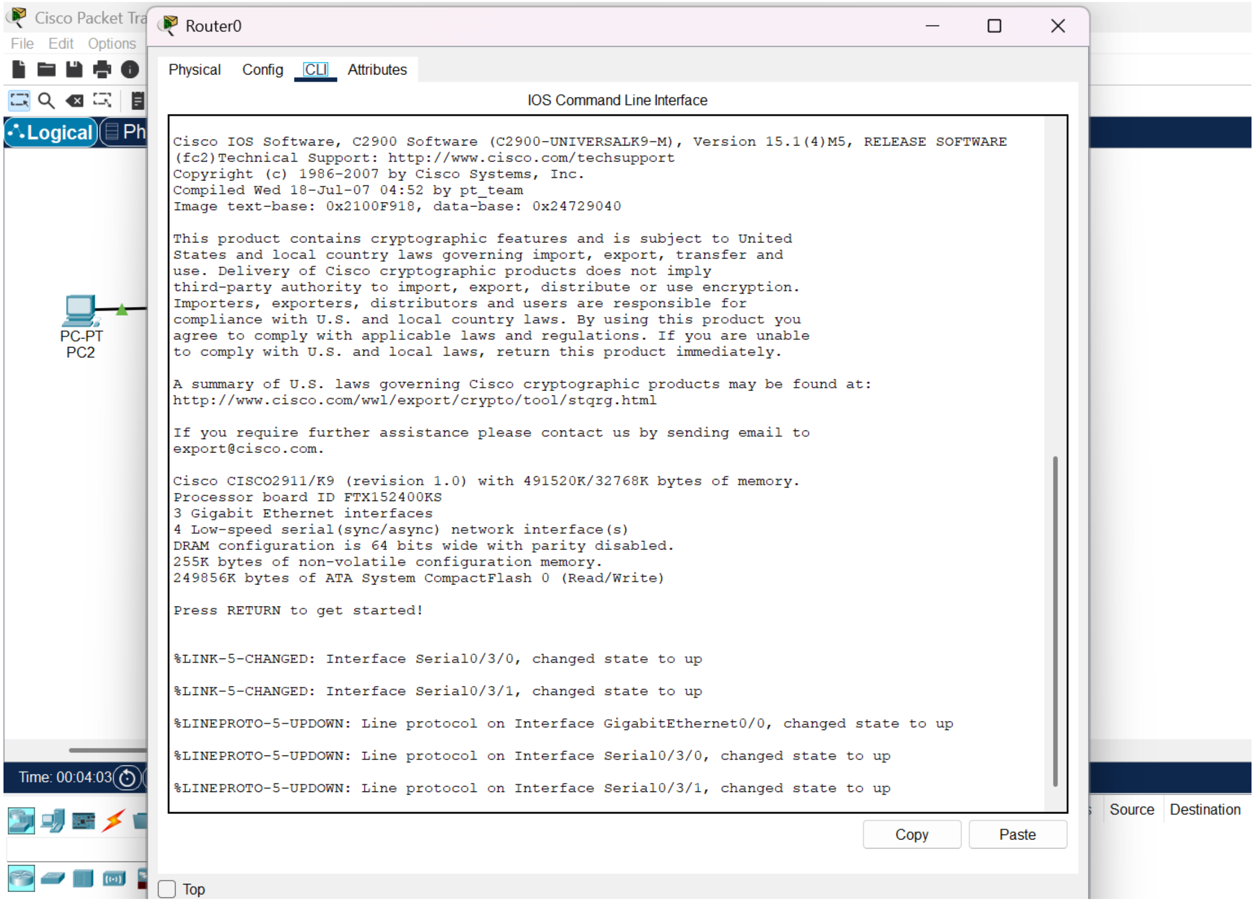


Figure 4: Router 0 CLI

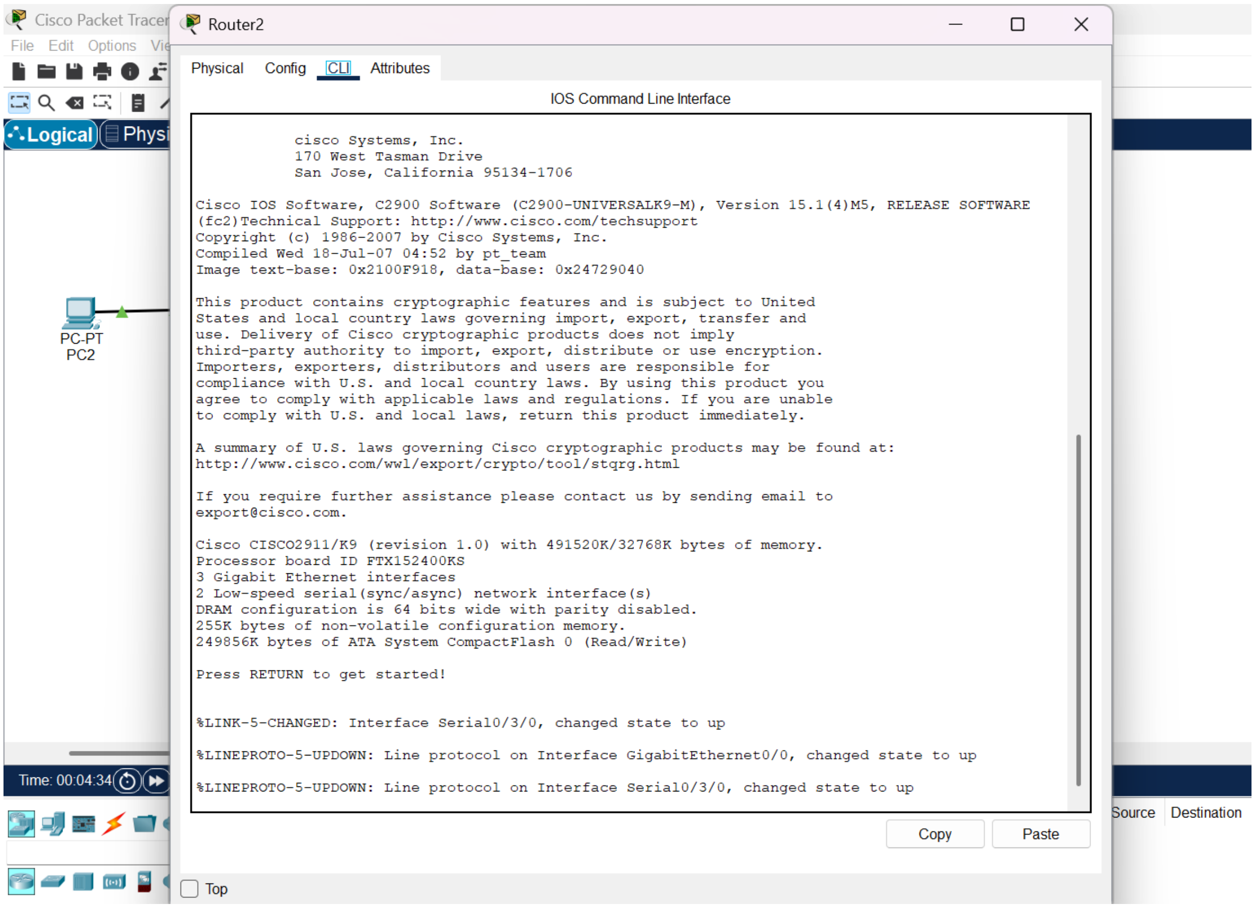


Figure 5: Router 2 CLI

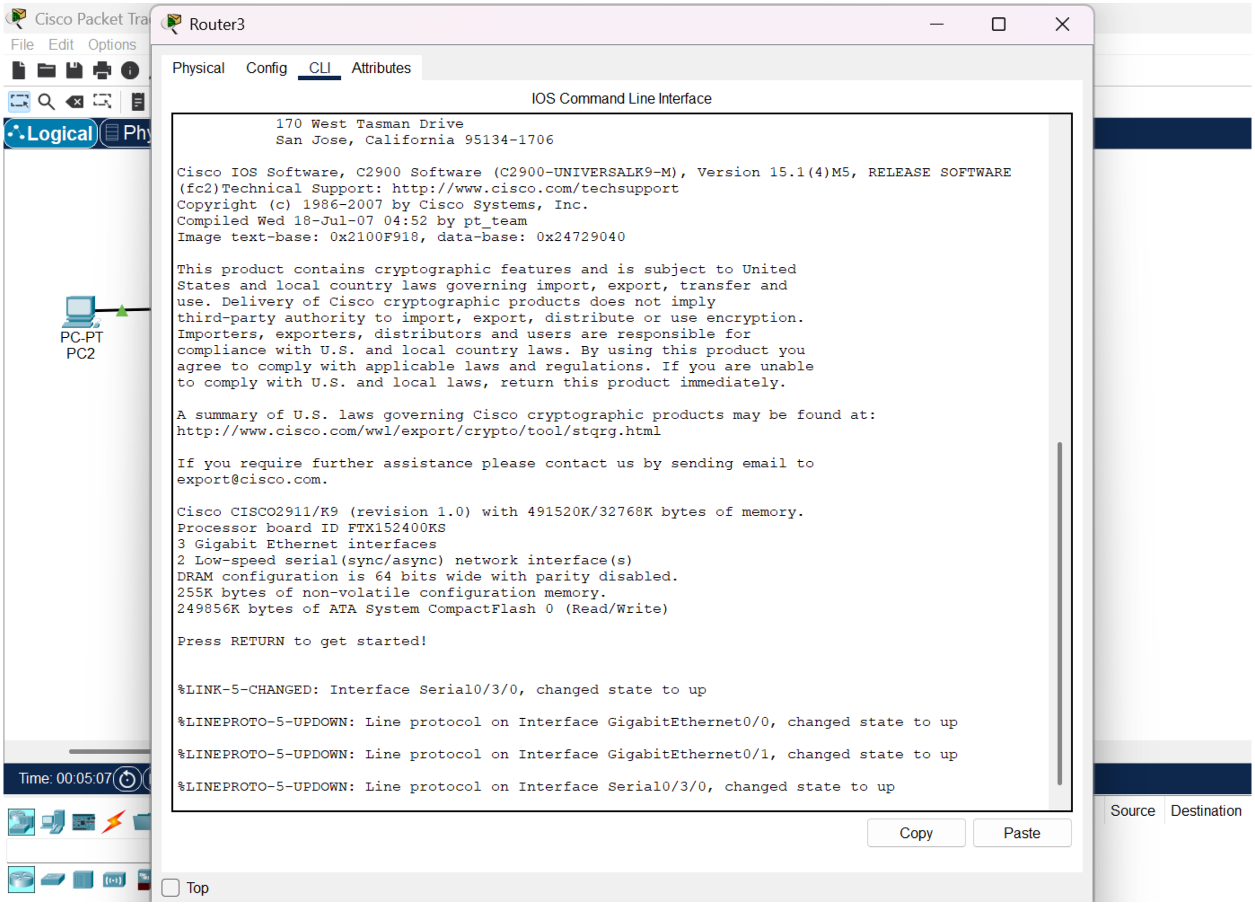


Figure 6: Router 3 CLI

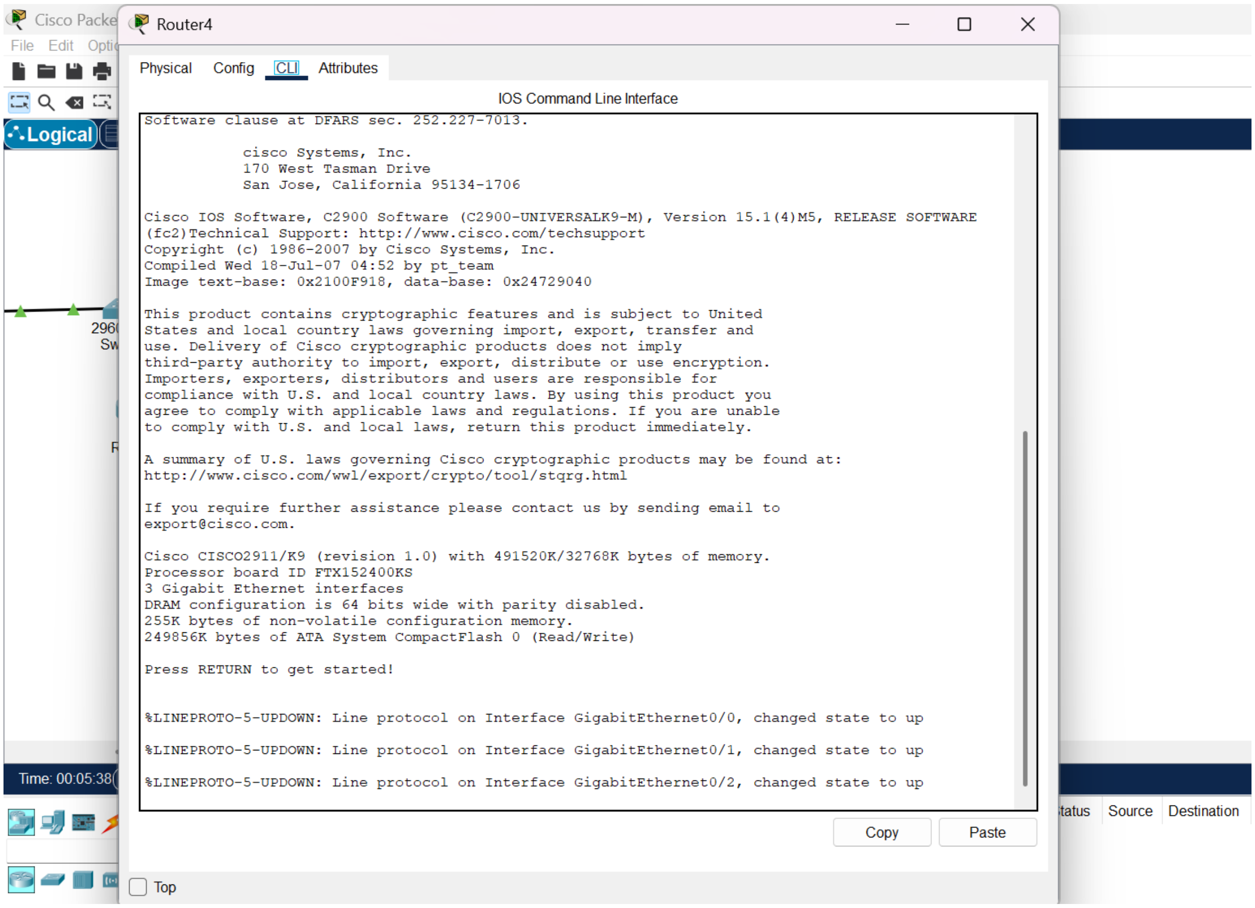


Figure 7: Router 4 CLI

2. IPv6 Addressing Scheme

Define the IPv6 addressing scheme for the network. Assign IPv6 addresses to interfaces, ensuring uniqueness within each network segment:

- **Router 0 to Router 1:** 2001:a::1/64
- **Router 0 to Router 2:** 2001:b::1/64
- **Router 1 to Router 3:** 2001:c::1/64
- **Router 2 to Router 4:** 2001:d::1/64
- **Router 3 to Router 4:** 2001:e::1/64
- **LAN 1 PC2:** 3001:a::1/64
- **LAN 2:** 3001:b::1/64
- **LAN 3:** 3001:c::1/64

3. Command to Add IPv6 Addresses

The command "ip address ipv6-address/prefix-length eui-64" is utilized to configure interfaces using the EUI-64 format. This format generates an IPv6 Global Unicast Address by incorporating the MAC address into the interface identifier.

4. IPv6 Configuration in LAN PCs

Configure IPv6 addresses for PCs in each LAN:

- **LAN 01 PC 02:**
    - IPv6 Address: 3001:a::2/64
    - Gateway Address: 3001:a::1

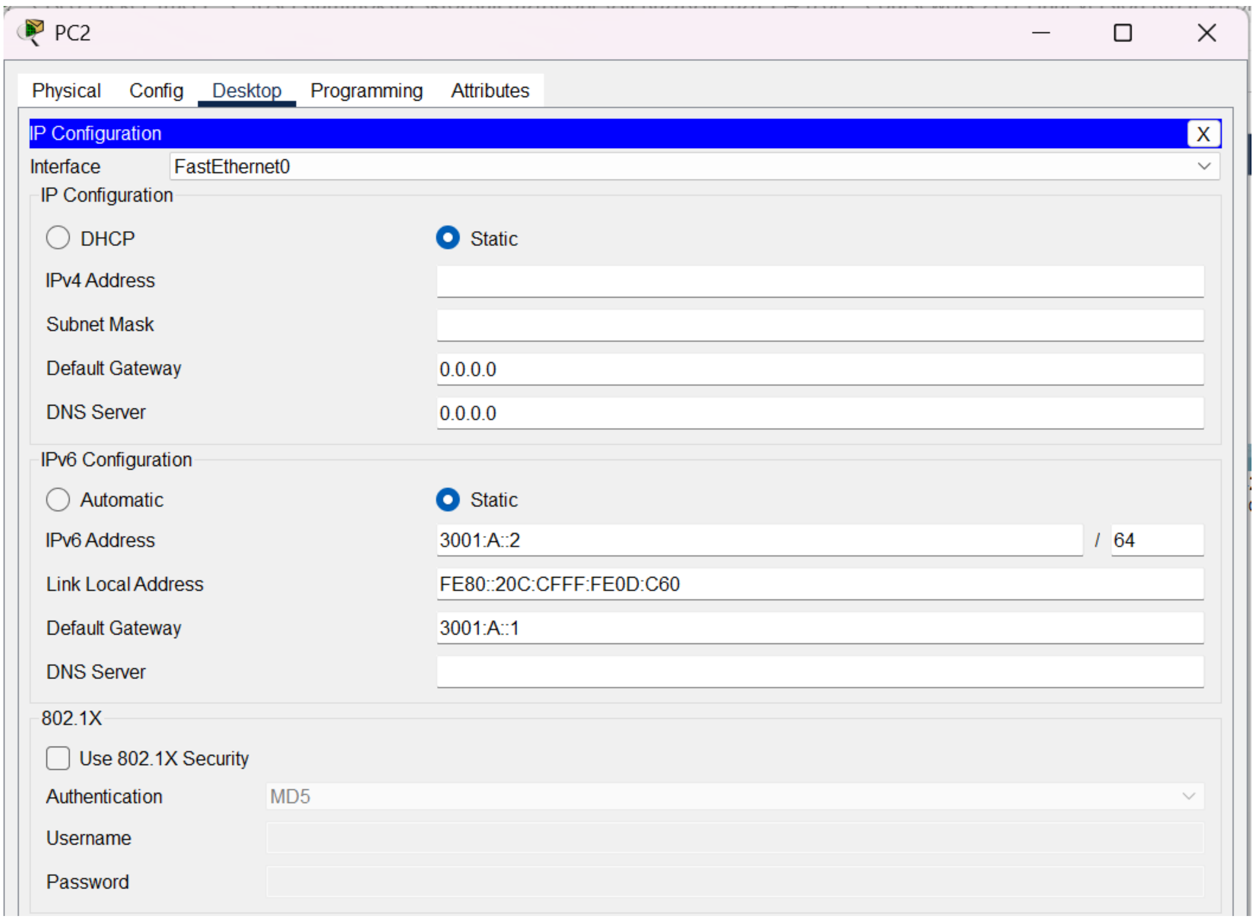


Figure 8: PC2 ->Desktop->IP config

- **LAN 02 PC0:**
    - IPv6 Address: 3001:b::2/64
    - Gateway Address: 3001:B::1

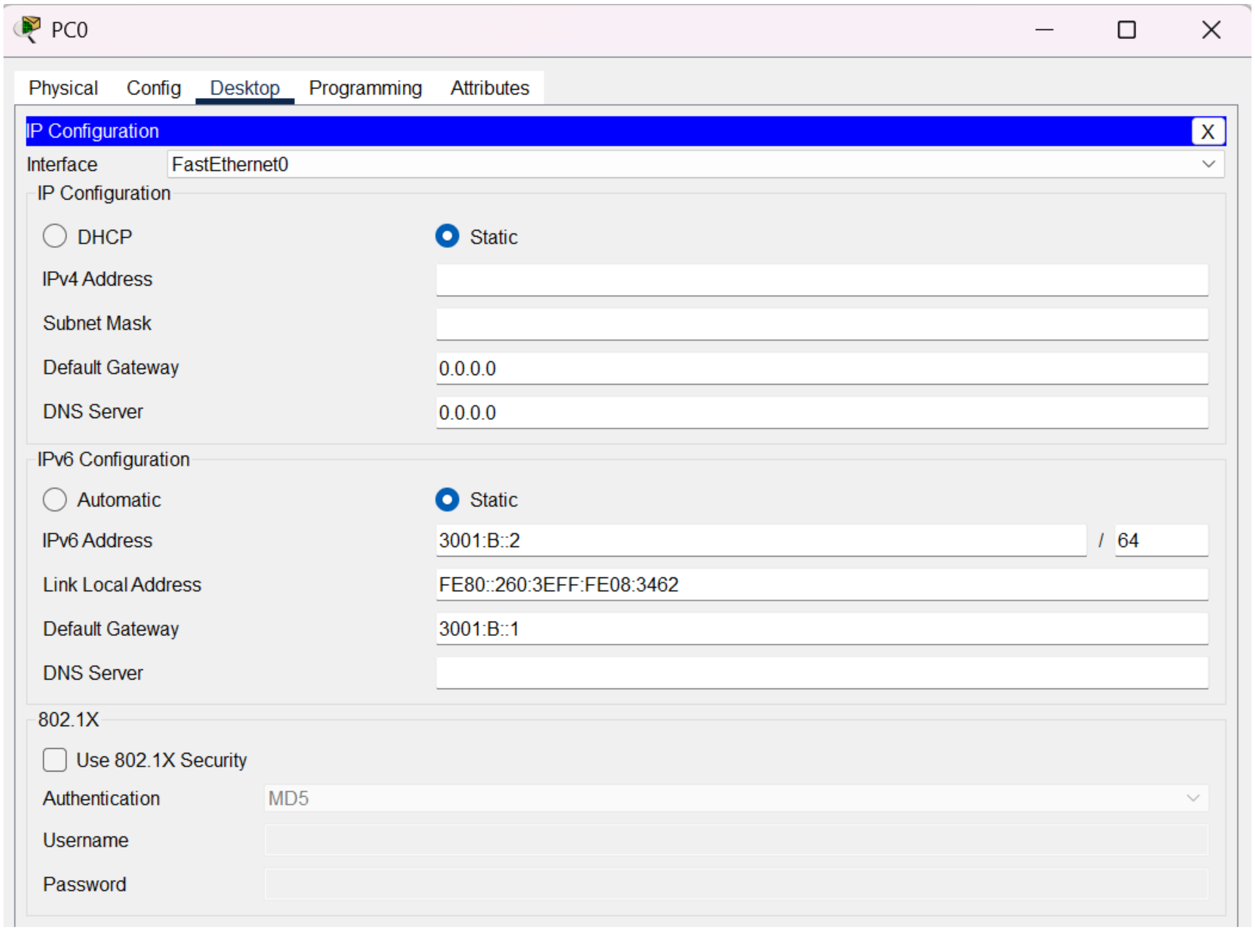


Figure 9: PC0 ->Desktop->IP config

- **LAN 03 PC 01:**
    - IPv6 Address: 3001:c::2/64
    - Gateway Address: 3001:c::1

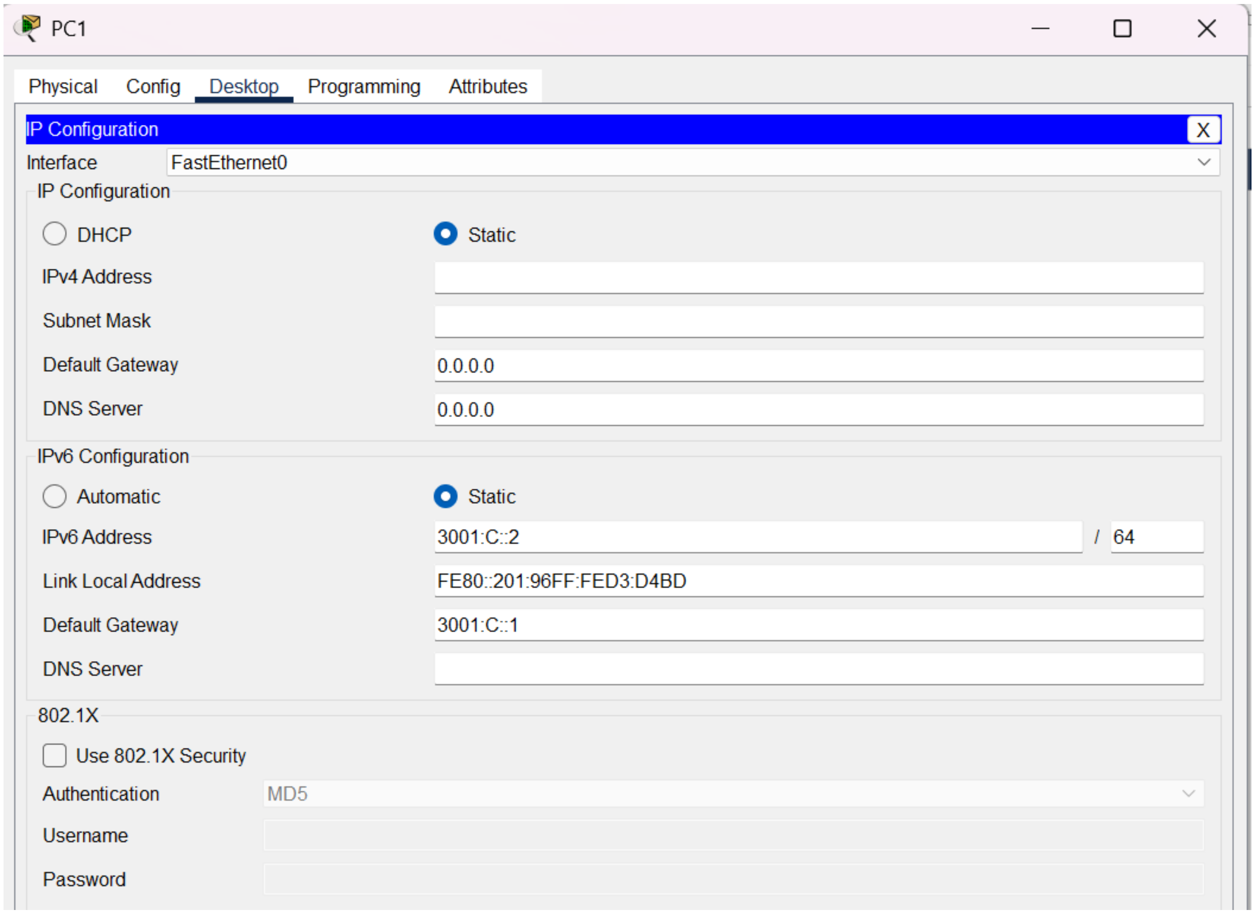


Figure 10: PC1 ->Desktop->IP config

5. Pinging Results from PCs to Local Gateway

Use the ping command to test connectivity between PCs and their local gateways:

- **LAN 01 PC:** Ping to Local Gateway (3001:a::1)

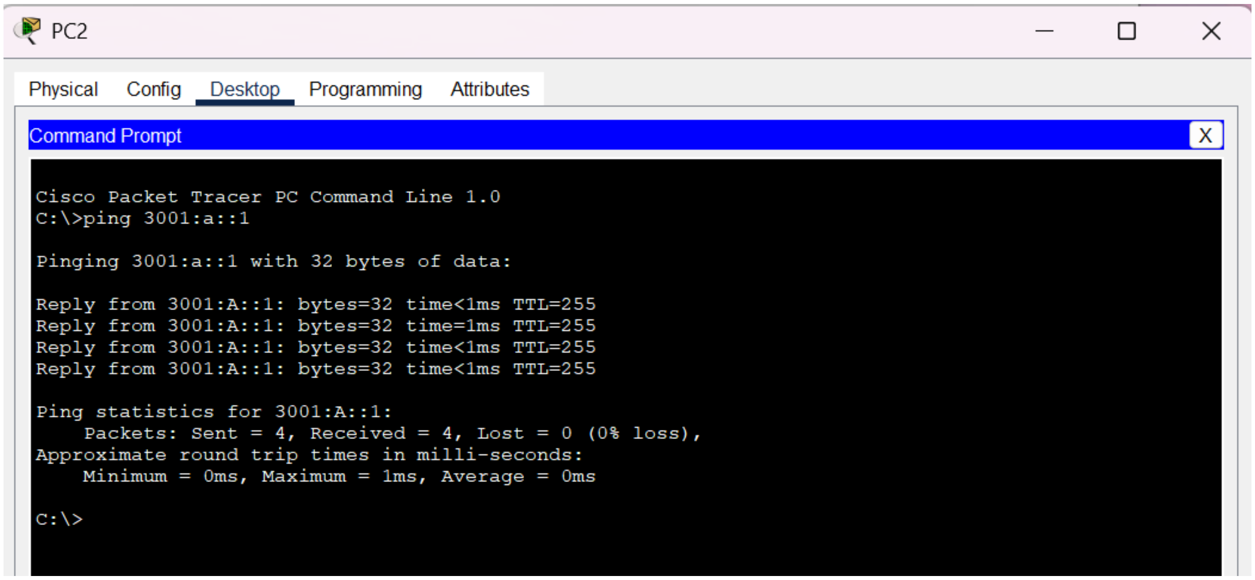


Figure 11: PC2-> Desktop-> Command Prompt

- **LAN 02 PC:** Ping to Local Gateway (3001:b::1)

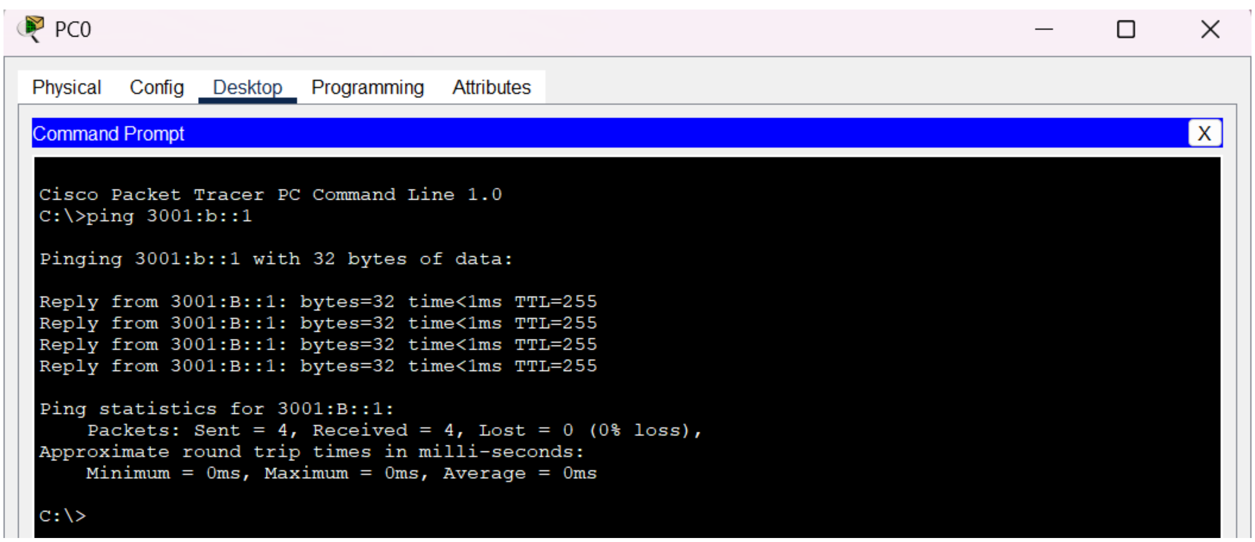


Figure 12: PC0-> Desktop-> Command Prompt

- **LAN 03 PC:** Ping to Local Gateway (3001:c::1)

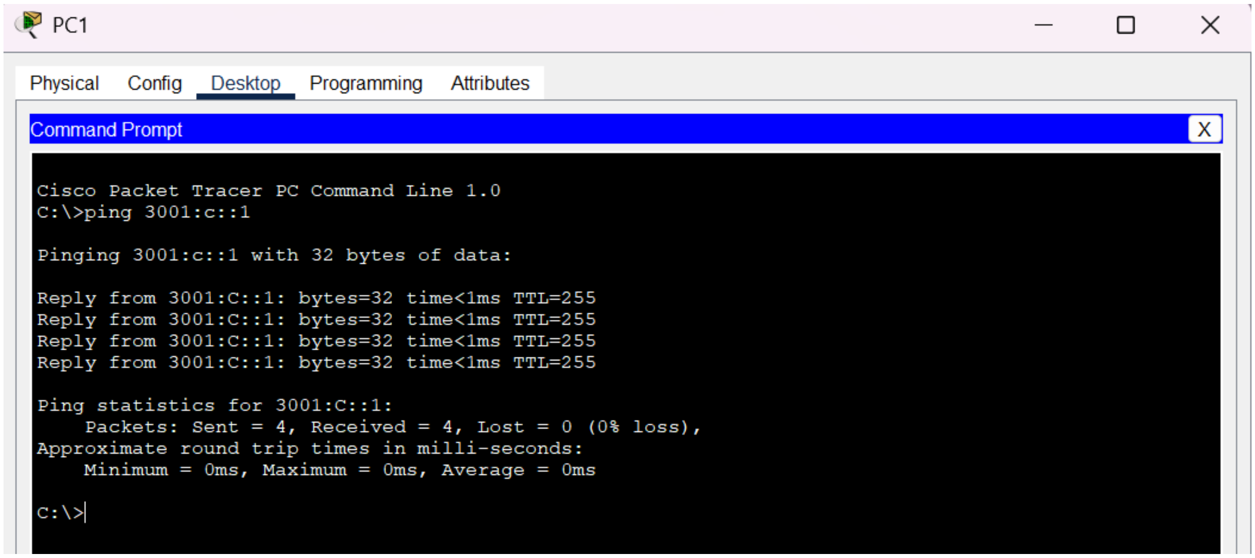
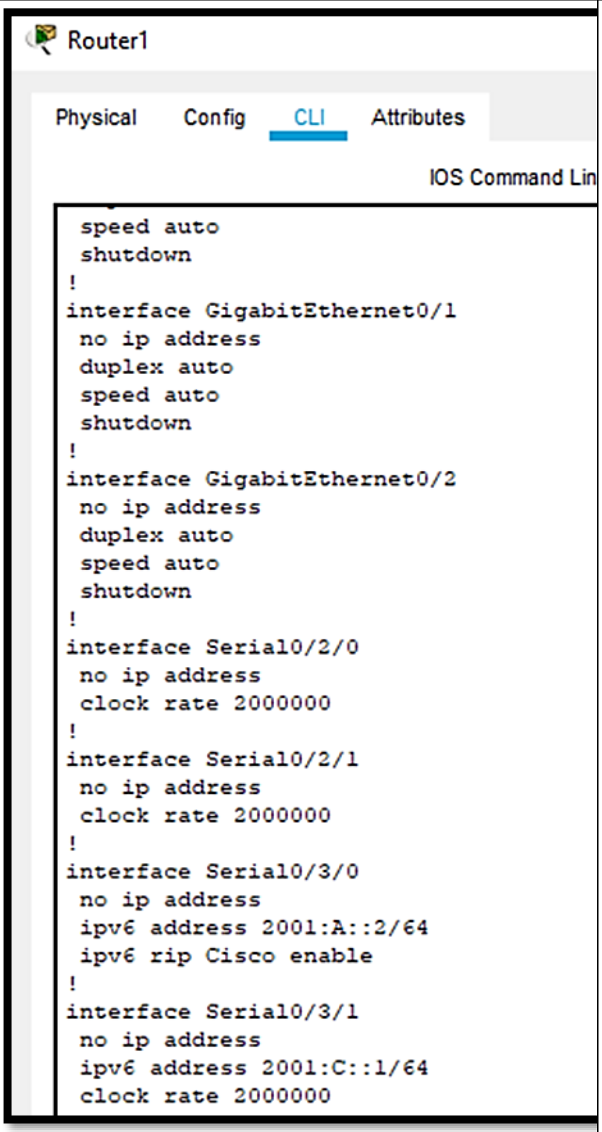


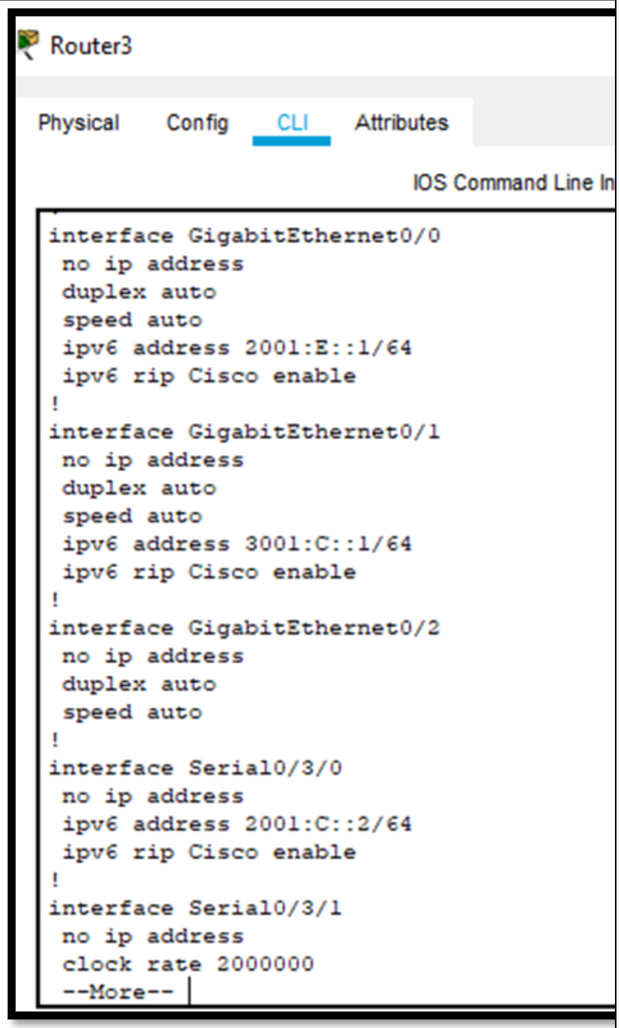
Figure 13: PC1-> Desktop-> Command Prompt

6. Configuring RIPng in Routers

Activate IPv6 on routers using the "ipv6 unicast-routing" command. Configure RIPng on each router to enable dynamic routing:

- Router 0 to Router 1: router ripng > network 2001:a::/64
- Router 0 to Router 2: router ripng > network 2001:b::/64
- Router 1 to Router 3: router ripng > network 2001:c::/64
- Router 2 to Router 4: router ripng > network 2001:d::/64
- Router 3 to Router 4: router ripng > network 2001:e::/64





7. IPv6 Routes Verification in Routers

Check the RIPng routing table in each router to verify the presence of IPv6 routes:

- **Router 0: show ipv6 route**

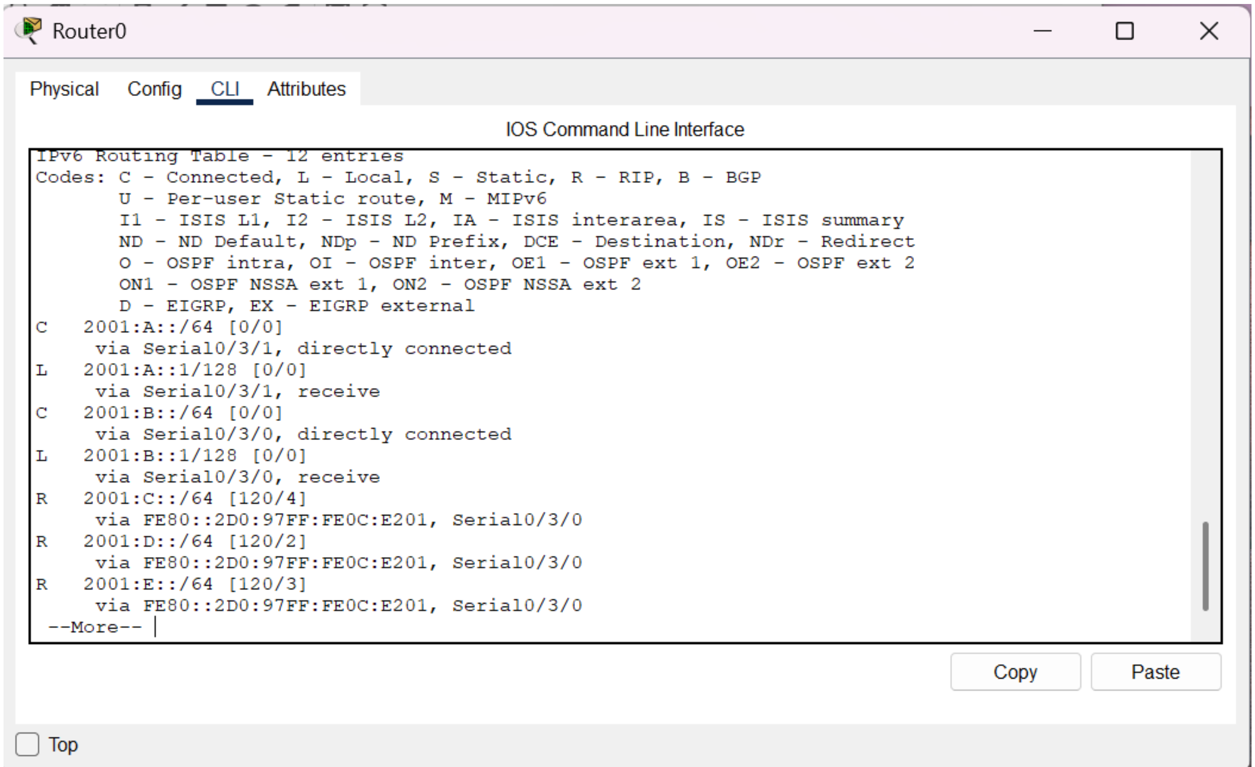


Figure 14: Router 0-- show ipv6 route

- **Router 1: show ipv6 route**

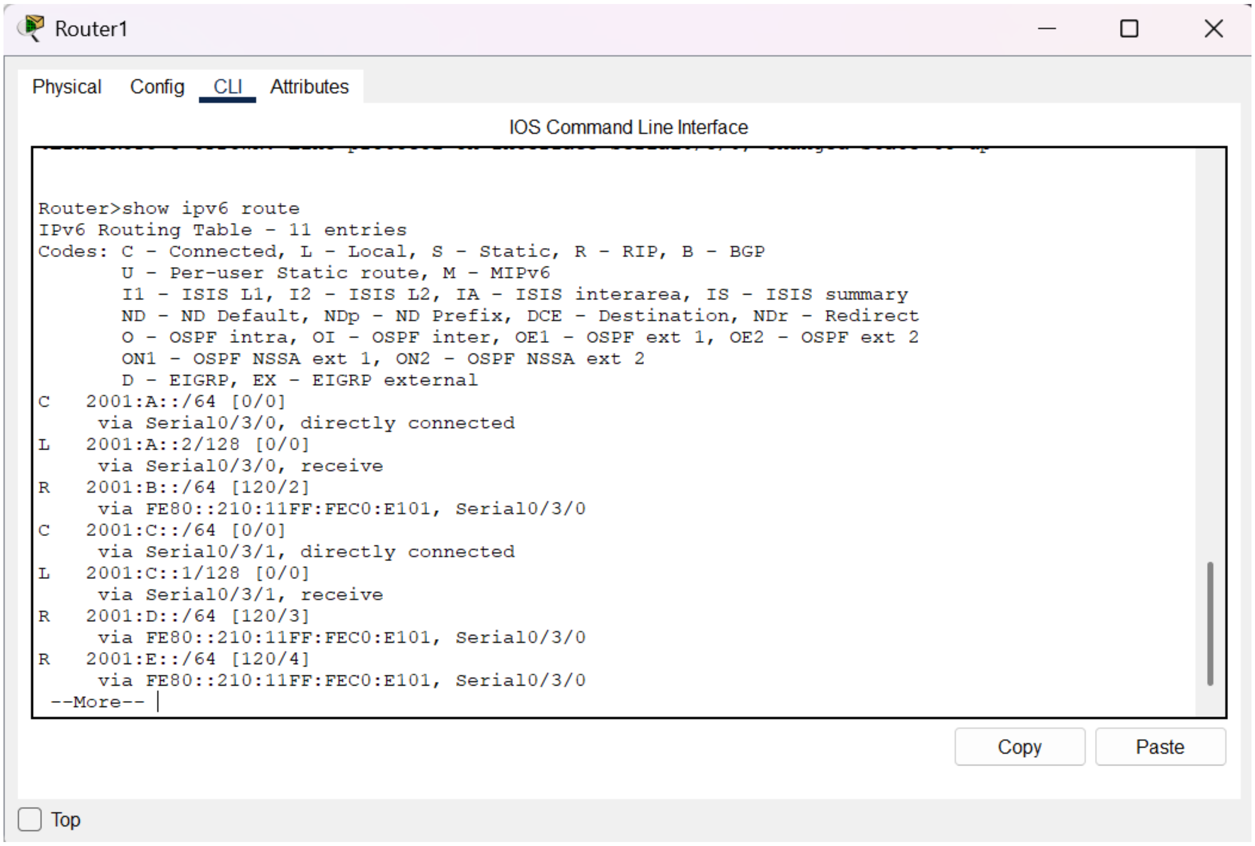


Figure 15: Router 1-- show ipv6 route

- **Router 3: show ipv6 route**

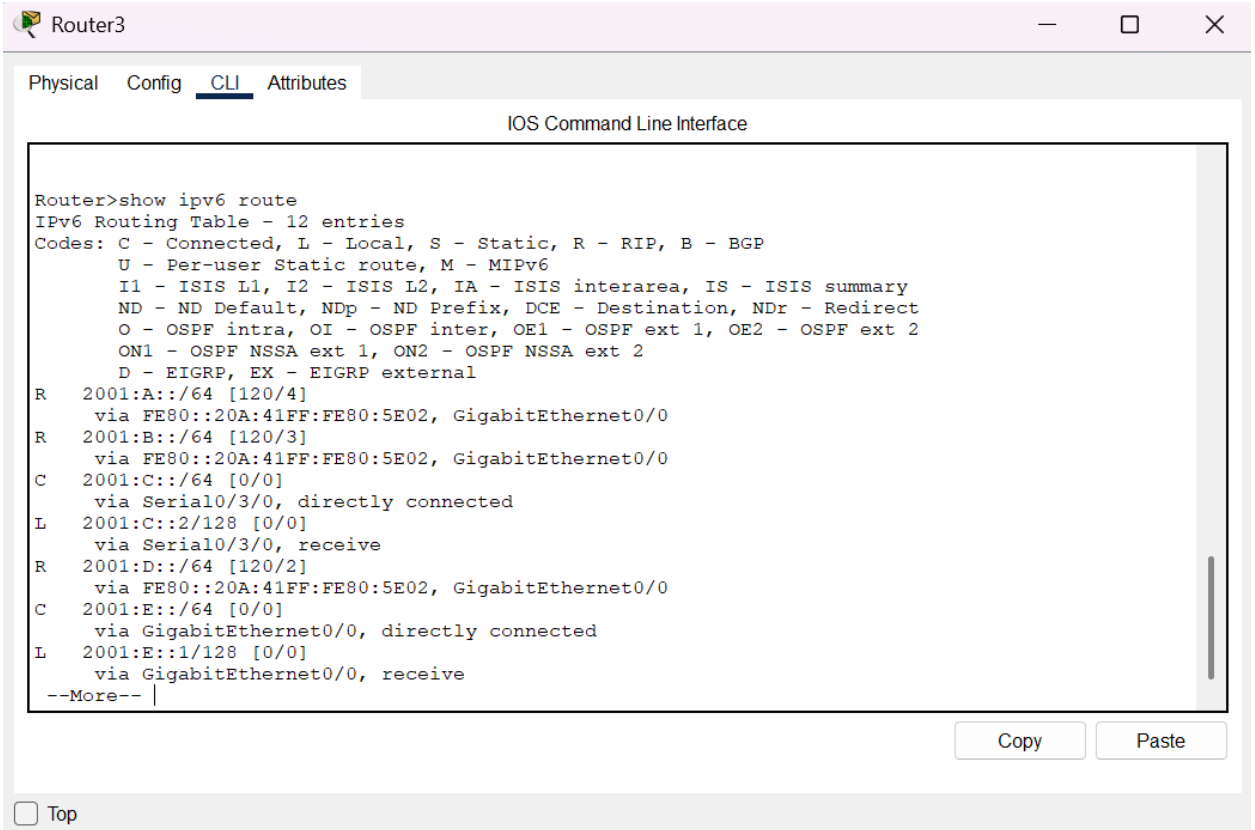


Figure 16: Router 2-- show ipv6 route

- **Router 4: show ipv6 route**

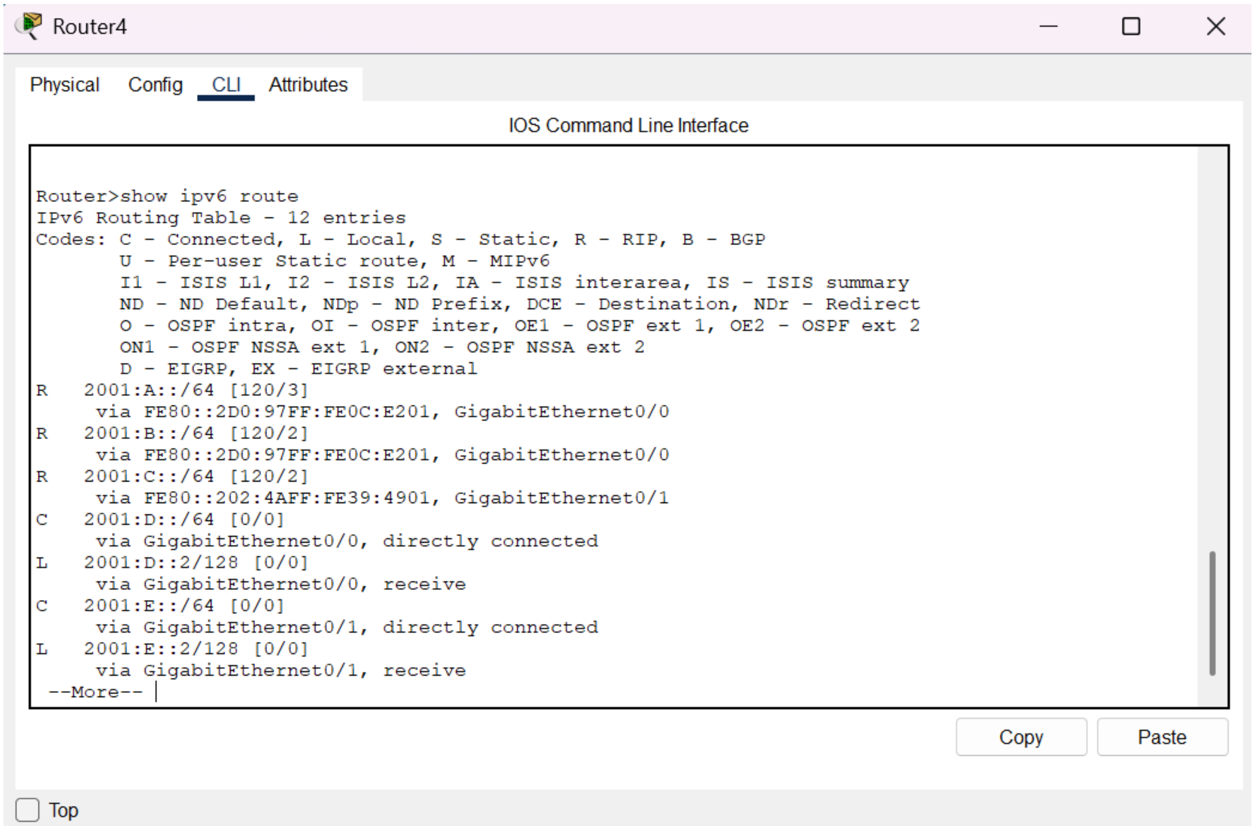


Figure 17: Router 3-- show ipv6 route

8. Pinging Results from LANs

Finally, test connectivity across the network using the ping command:

- LAN 1 to LAN 2: ping 3001:b::1
- LAN 1 to LAN 3: ping 3001:c::2
- LAN 2 to LAN 3: ping 3001:c::1

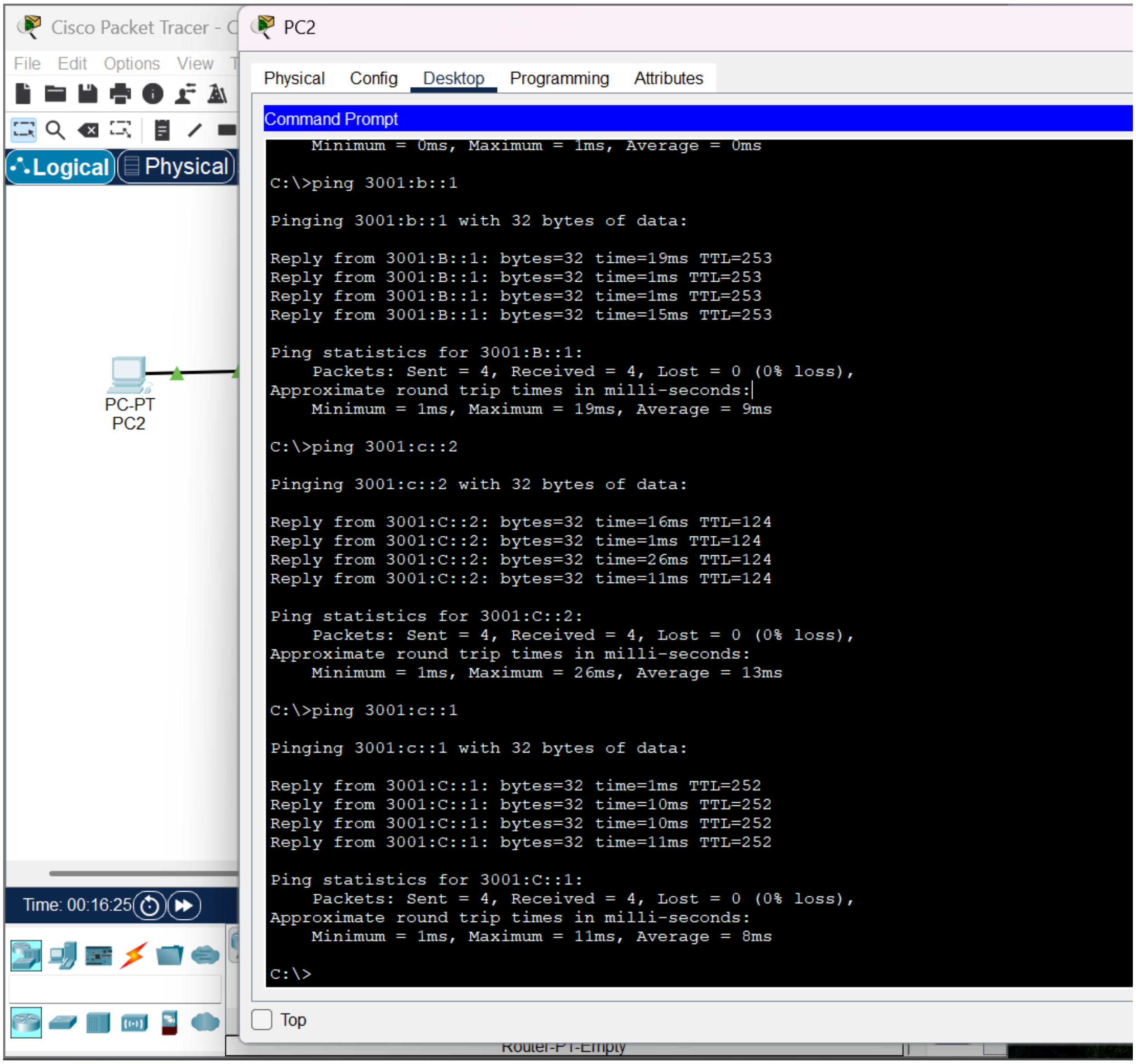


Figure 18: PC2->Desktop->Command Prompt-> ping commands

Detailed Configuration Explanation

IPv6 Address Format and EUI-64

IPv6 addresses, comprising 128 bits, are structured in eight groups of 16 bits each, separated by colons. The EUI-64 format is used for generating IPv6 Global Unicast Addresses, incorporating the MAC address into the interface identifier. This is done through the command "ip address ipv6-address/prefix-length eui-64."

IPv6 Addressing Scheme

The chosen addressing scheme ensures uniqueness and efficient routing within the network. Each router-to-router and LAN segment has a defined IPv6 address range, facilitating proper network segmentation.

RIPng Configuration

RIPng is configured on each router using the router ripng command and specifying the network to be advertised. This dynamic routing protocol ensures that routers exchange routing information, allowing for optimal path selection.

Verification and Testing

Verification steps include checking the IPv6 routes in each router to ensure RIPng has successfully advertised routes. Additionally, ping tests from LANs to their local gateways and across different LANs confirm end-to-end connectivity.

3. OSPFv3 (Single Area) Implementation

Introduction to OSPFv3

The Open Shortest Path First (OSPF) link-state routing protocol, specifically OSPFv3, employs a shortest path first algorithm to determine the most efficient route within a network. Developed by the Internet Engineering Task Force (IETF), OSPFv3 enhances packet transmission in substantial autonomous systems or routing domains. This protocol, operating on port 89 with an Administrative Distance (AD) value of 110, is classified as a network layer protocol. The multicast addresses 224.0.0.5 and 224.0.0.6 are utilized for standard OSPF communication and updates related to the Designated Router (DR) and Backup Designated Router (BDR), respectively.

IPv6 Addressing in OSPFv3

In OSPFv3, IPv6 addressing semantics have been integrated seamlessly into the protocol. Significant shifts include the elimination of IPv6 addressing from OSPF packets and primary Link State Advertisement (LSA) types to achieve protocol independence. Notable changes involve the absence of party addresses in "hello" packets, the utilization of the EUI-64 format for the interface identifier, and the expression of LSA addresses in prefix format.

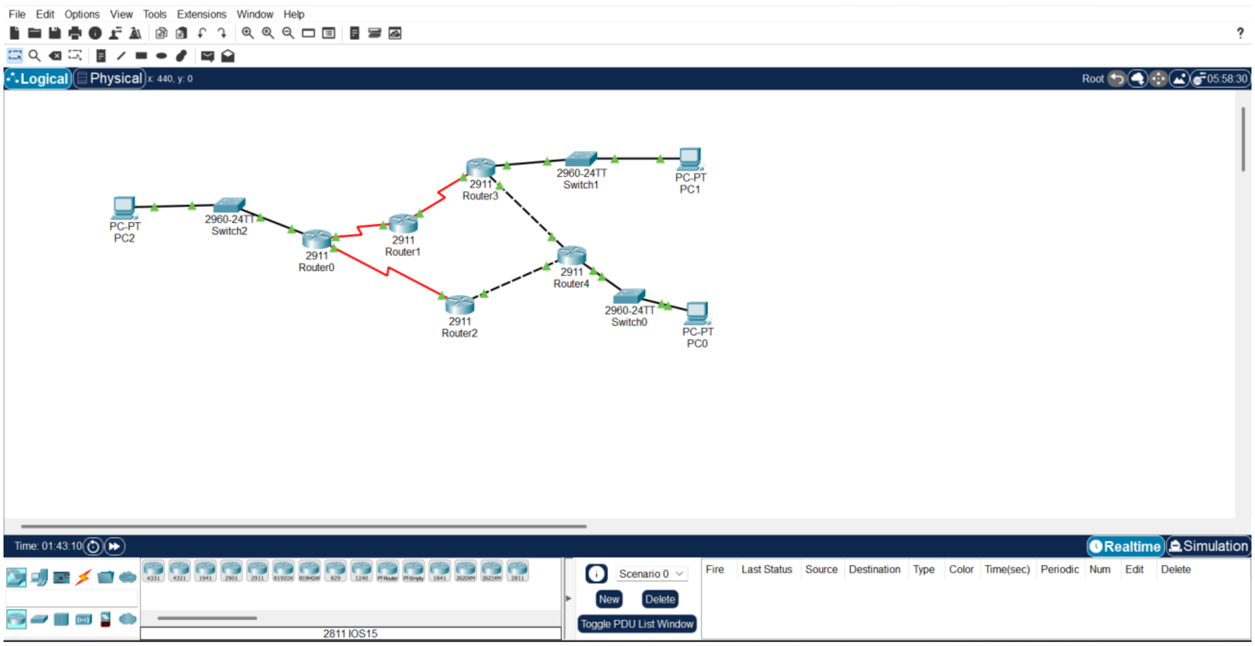


Figure 19: cisco Workspace

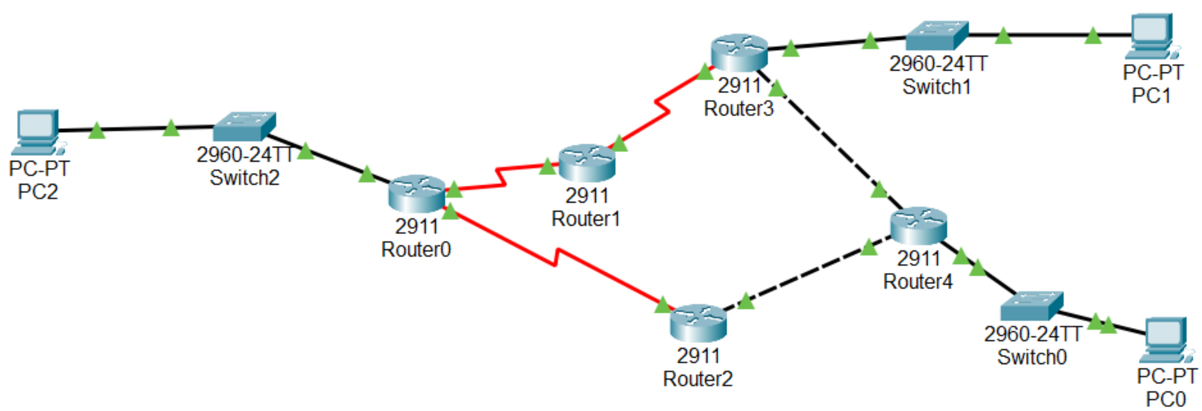


Figure 20: Final Network Topology

Topology Configuration Steps

IPv6 Addressing Table

Before delving into the OSPFv3 implementation details, let's review the IPv6 addressing scheme for the network:

- **Router 0 to Router 1:** 2001:a::1/64
- **Router 0 to Router 2:** 2001:b::1/64
- **Router 1 to Router 3:** 2001:c::1/64
- **Router 2 to Router 4:** 2001:d::1/64

- **Router 3 to Router 4:** 2001:e::1/64
- **LAN 1:** 3001:a::1/64
- **LAN 2:** 3001:b::1/64
- **LAN 3:** 3001:c::1/64

OSPFv3 Implementation Steps

Let's go through the comprehensive steps to implement OSPFv3 in Packet Tracer:

Configuring IPv6 Addresses on Router Interfaces:

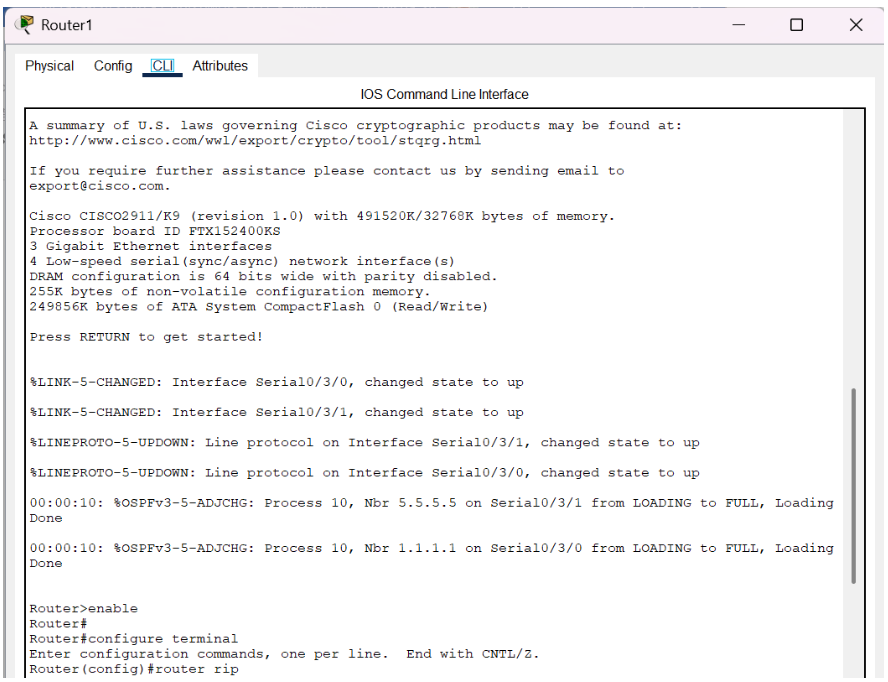


Figure 21: Router1 CLI

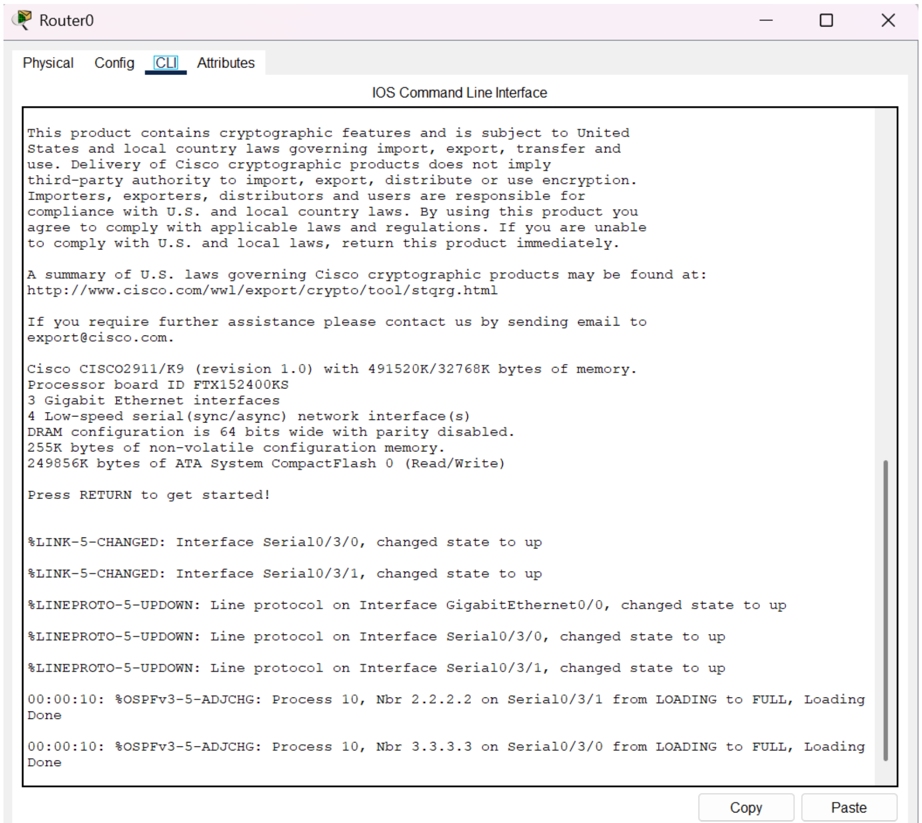


Figure 22: Router 0 CLI

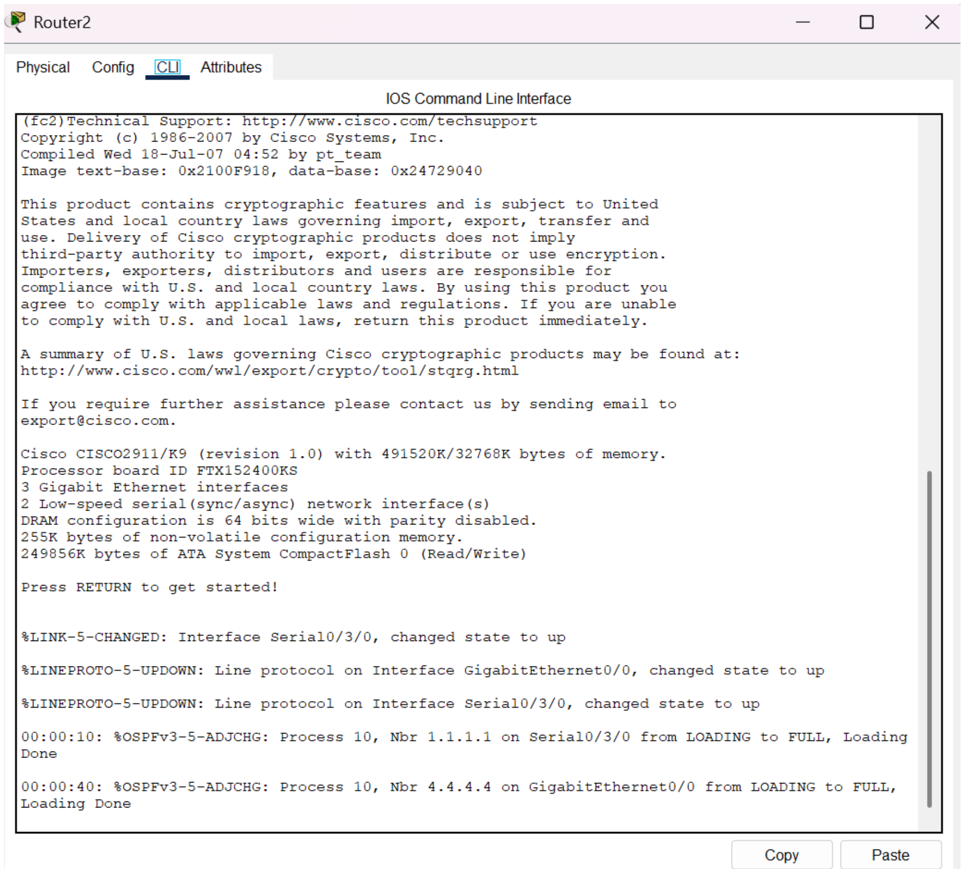


Figure 23: Router 2 CLI

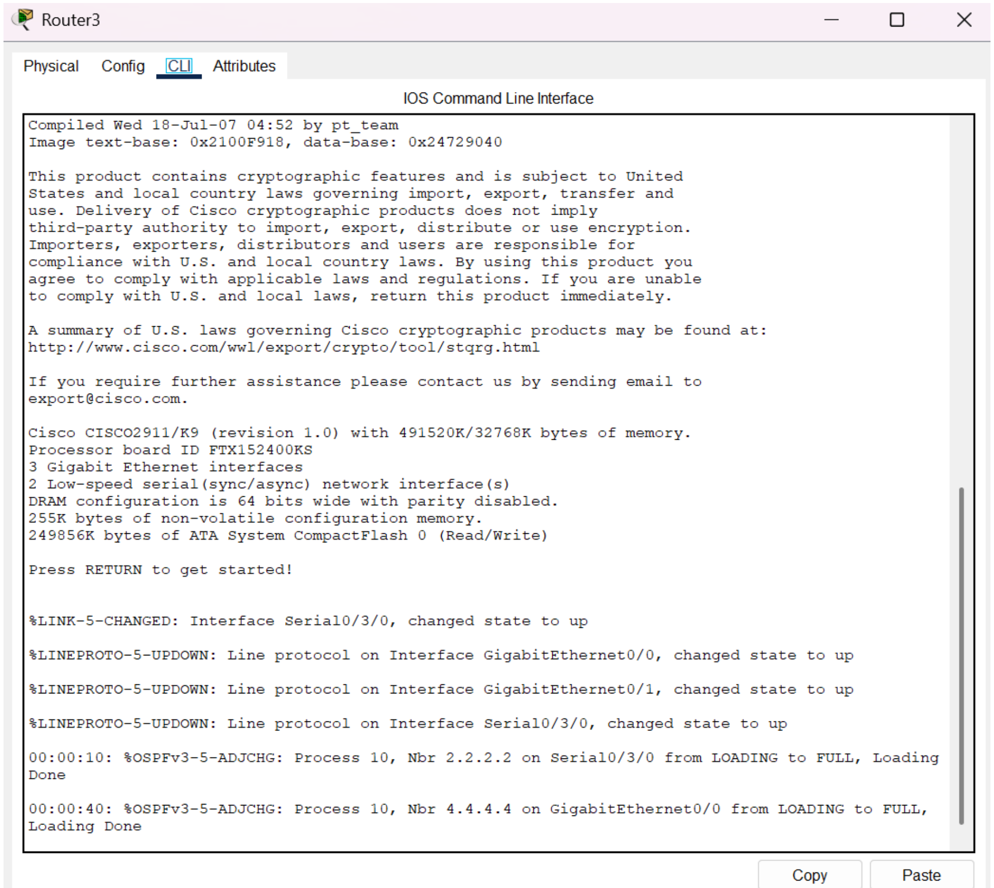


Figure 24: Router 3 CLI

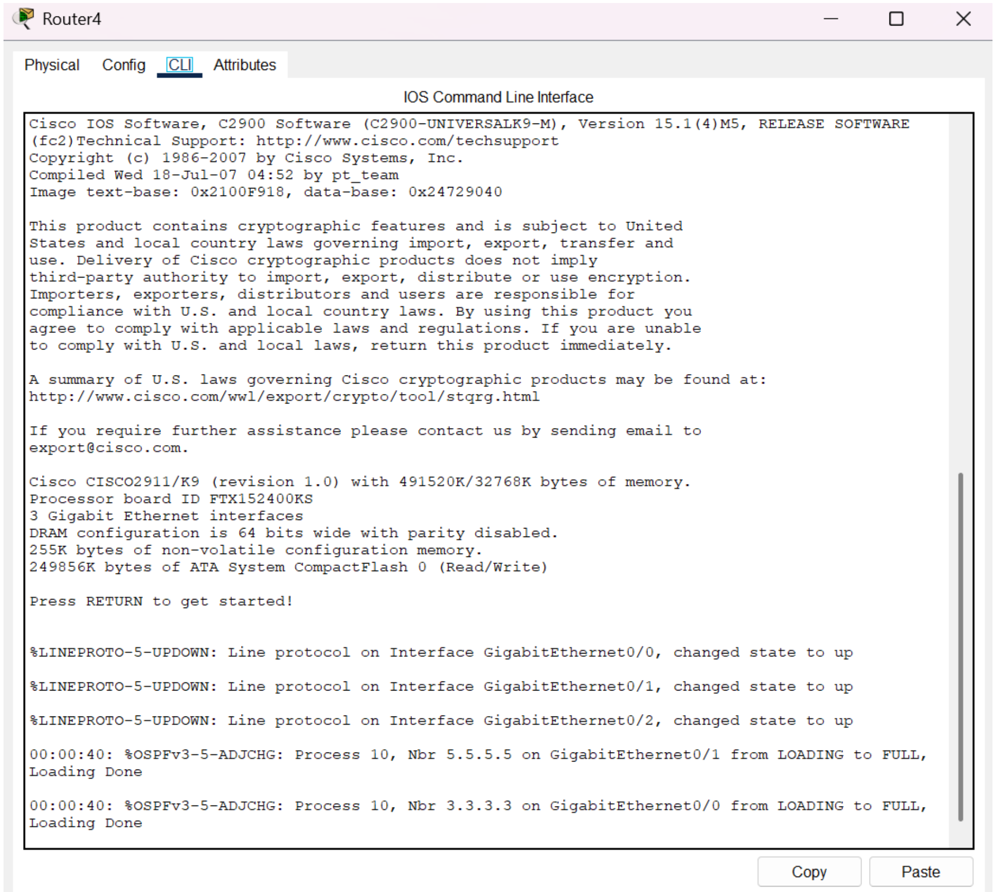


Figure 25: Router 4 CLI

Enable IPv6 Unicast Routing:

In global configuration mode, use the command ipv6 unicast-routing on both routers to enable IPv6 routing.

Interface IPv6 Configuration:

Assign IPv6 addresses to the interfaces of each router based on the addressing table.

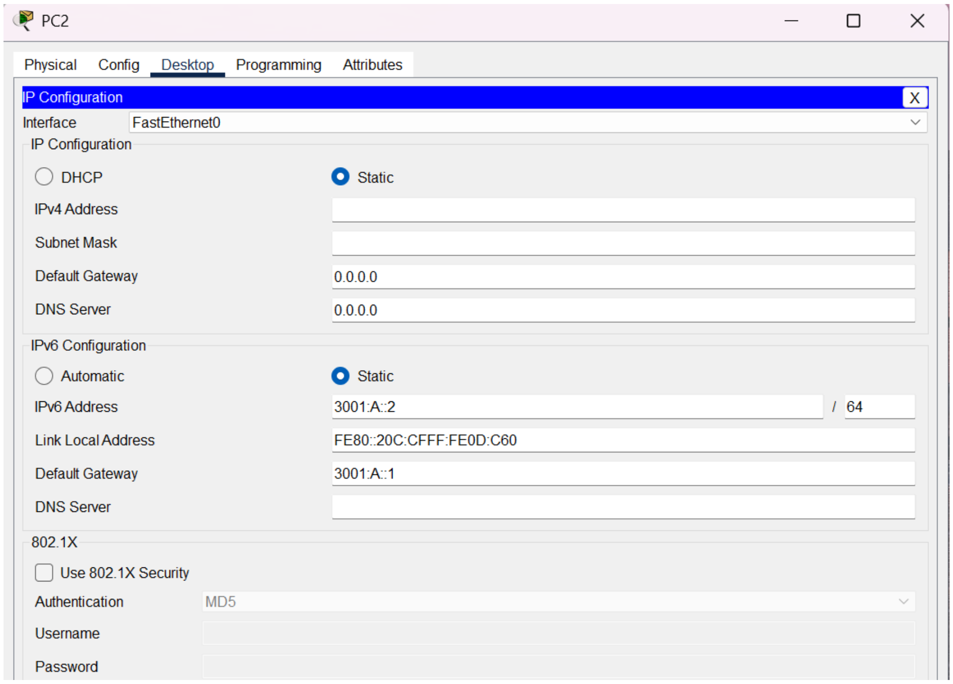


Figure 26: PC2 ->Desktop->IP config

- **LAN 02 PC0:**

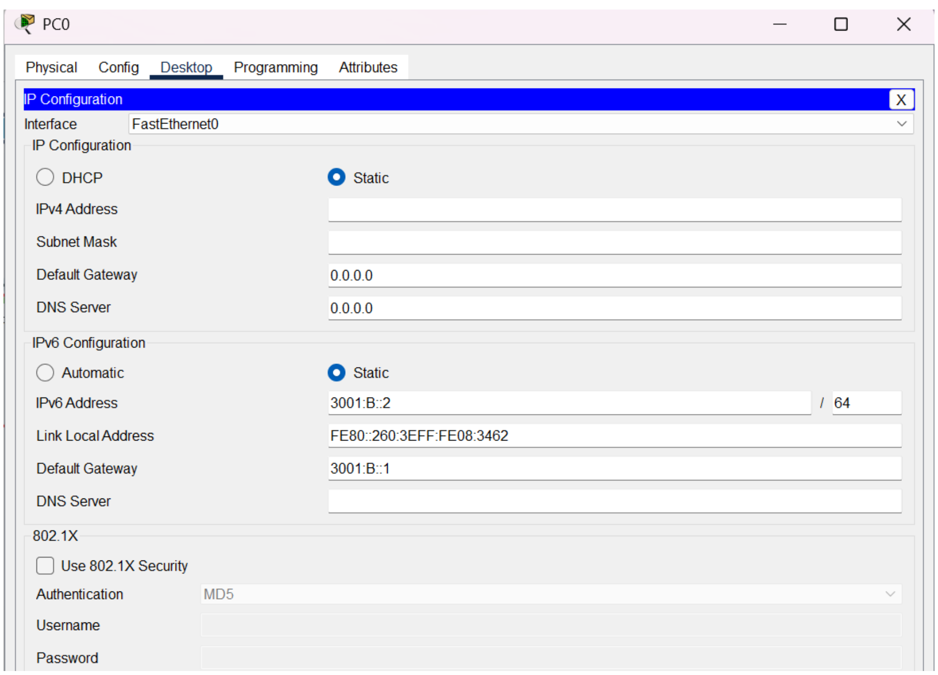


Figure 27: PC0 ->Desktop->IP config

- **LAN 03 PC 01:**

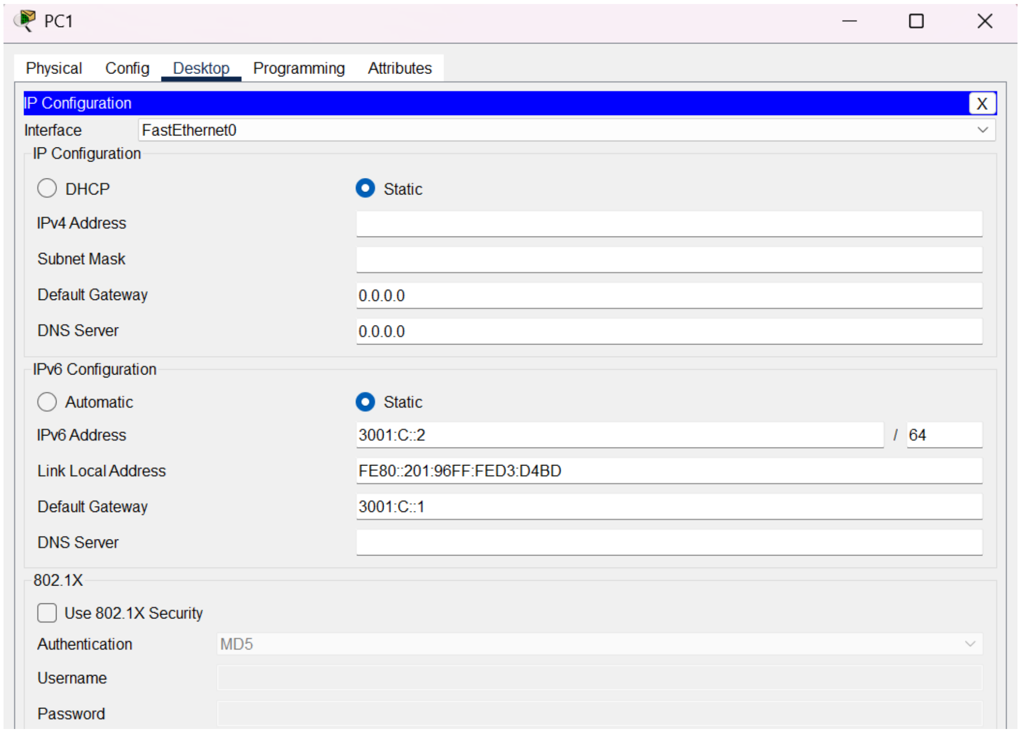


Figure 28: PC1 ->Desktop->IP config

OSPFv3 Configuration:

Enter OSPFv3 configuration mode using the command ipv6 router ospf [process-id]. Configure OSPFv3 on each router for the respective networks using the area [area-id] command.

IPv6 Configuration in LAN PCs:

Configure IPv6 addresses for PCs in each LAN segment along with the gateway address.

Use the ping command to test connectivity between PCs and their local gateways:

- **LAN 01 PC:** Ping to Local Gateway (3001:a::1)

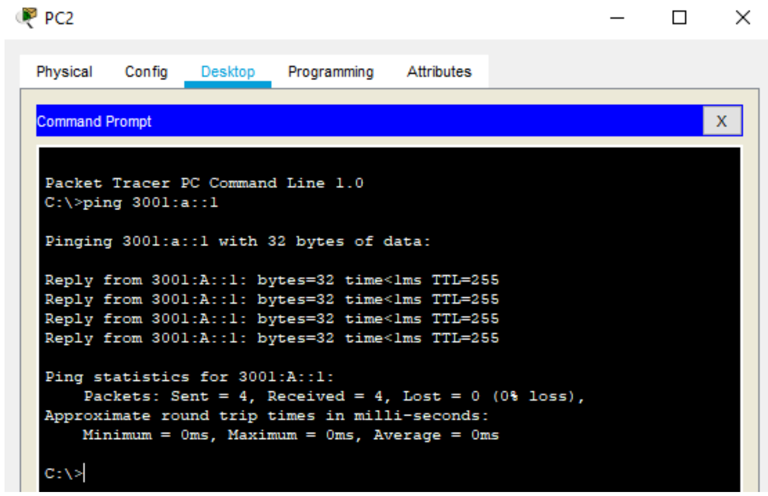


Figure 29: PC2-> Desktop-> Command Prompt

- **LAN 02 PC:** Ping to Local Gateway (3001:b::1)

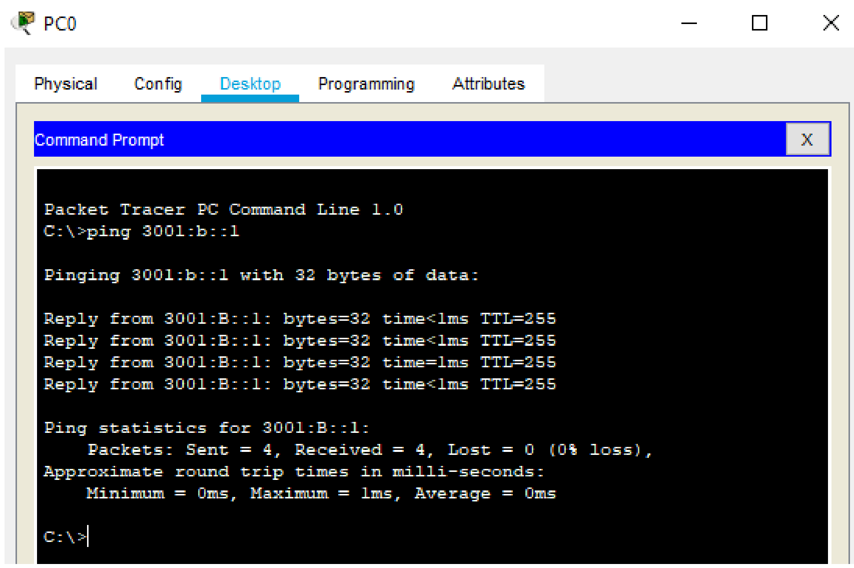


Figure 30: PC0-> Desktop-> Command Prompt

- **LAN 03 PC:** Ping to Local Gateway (3001:c::1)

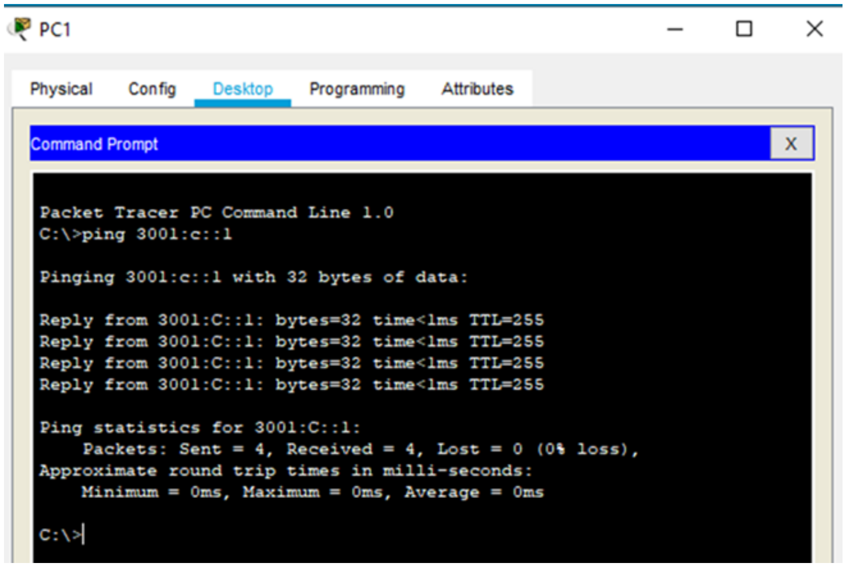
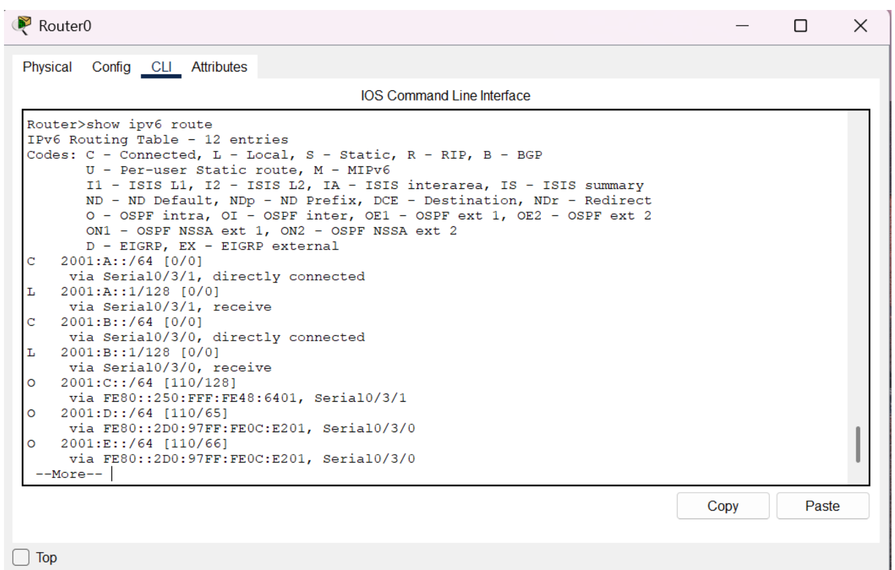


Figure 31: PC1-> Desktop-> Command Prompt

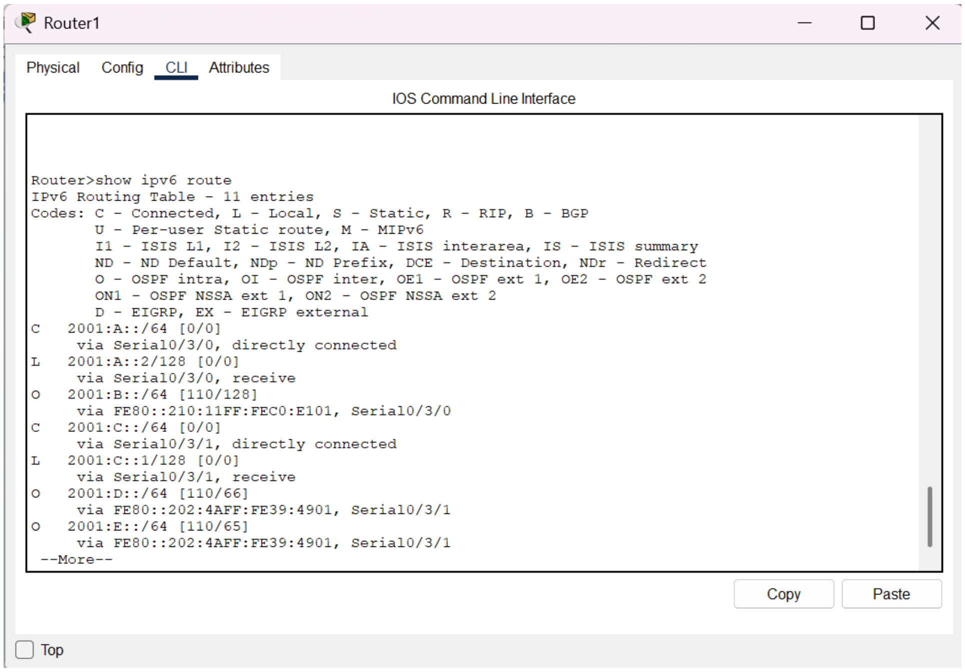
IPv6 Routes from Routers:

Check the RIPng routing table in each router to verify the presence of IPv6 routes:

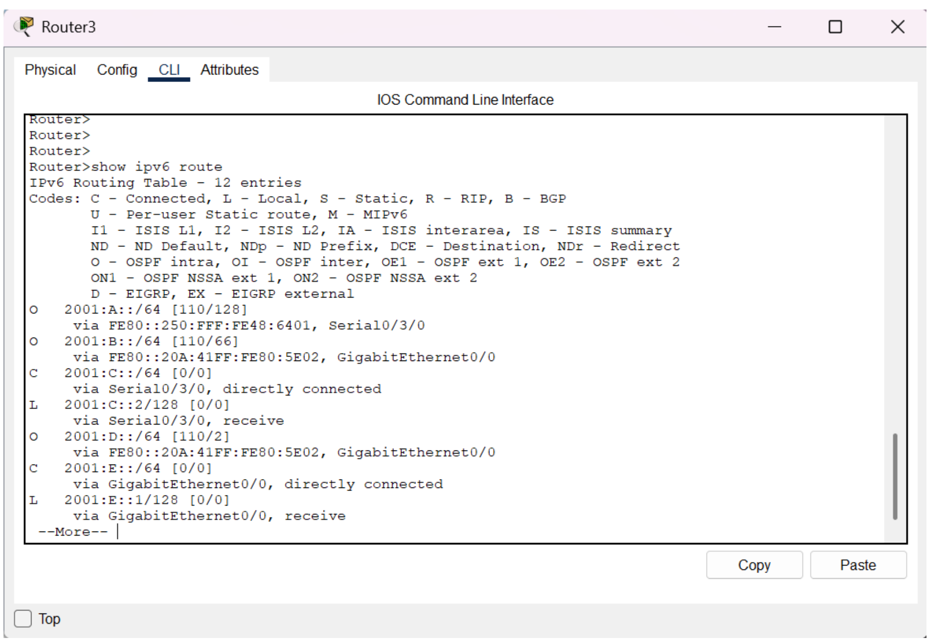
- **Router 0: show ipv6 route**



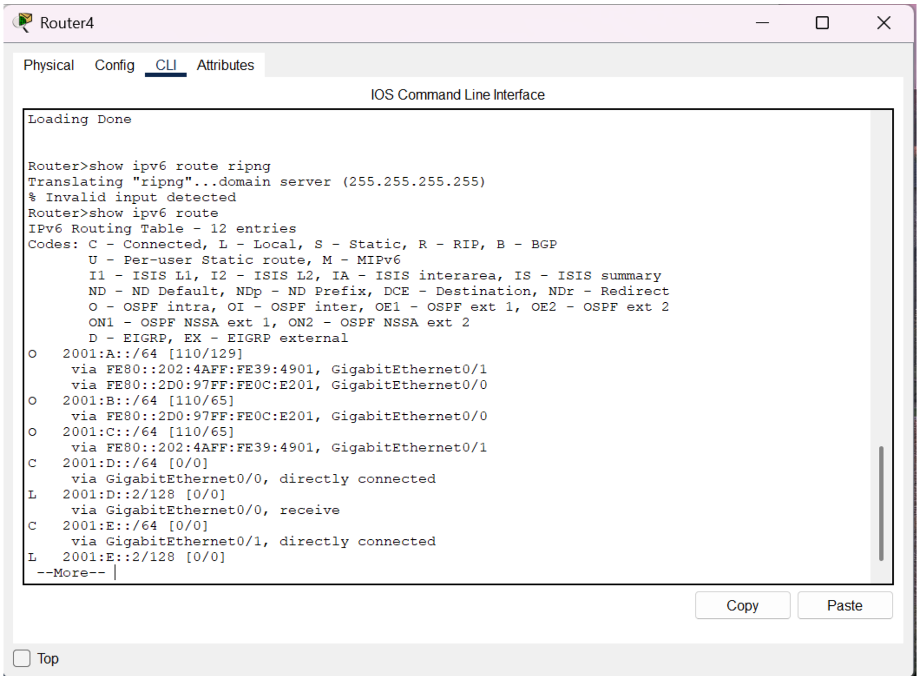
- **Router 1: show ipv6 route**



- **Router 3: show ipv6 route**



- **Router 4: show ipv6 route**

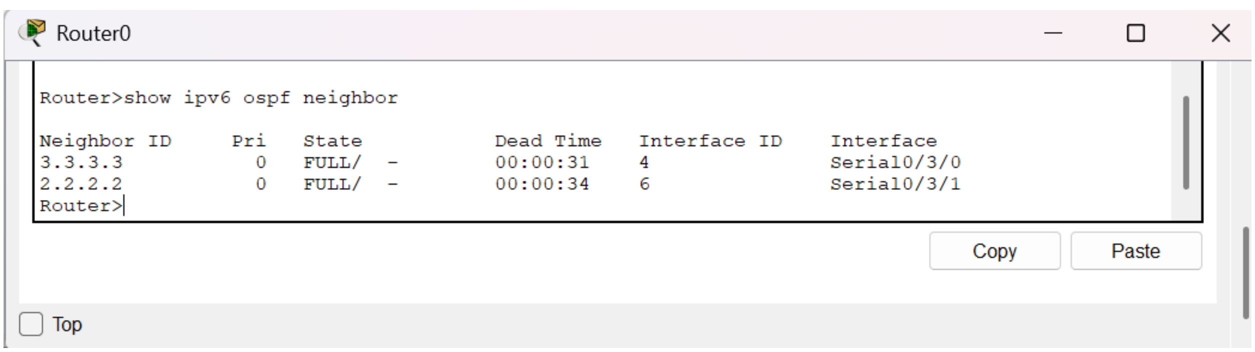



```
Loading Done


Router>show ipv6 route ripng
Translating "ripng"...domain server (255.255.255.255)
% Invalid input detected
Router>show ipv6 route
IPv6 Routing Table - 12 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
O   2001:A::/64 [110/129]
     via FE80::202:4AFF:FE39:4901, GigabitEthernet0/1
     via FE80::2D0:97FF:FE0C:E201, GigabitEthernet0/0
O   2001:B::/64 [110/65]
     via FE80::2D0:97FF:FE0C:E201, GigabitEthernet0/0
O   2001:C::/64 [110/65]
     via FE80::202:4AFF:FE39:4901, GigabitEthernet0/1
C   2001:D::/64 [0/0]
     via GigabitEthernet0/0, directly connected
L   2001:D::2/128 [0/0]
     via GigabitEthernet0/0, receive
C   2001:E::/64 [0/0]
     via GigabitEthernet0/1, directly connected
L   2001:E::2/128 [0/0]
 --More--
```

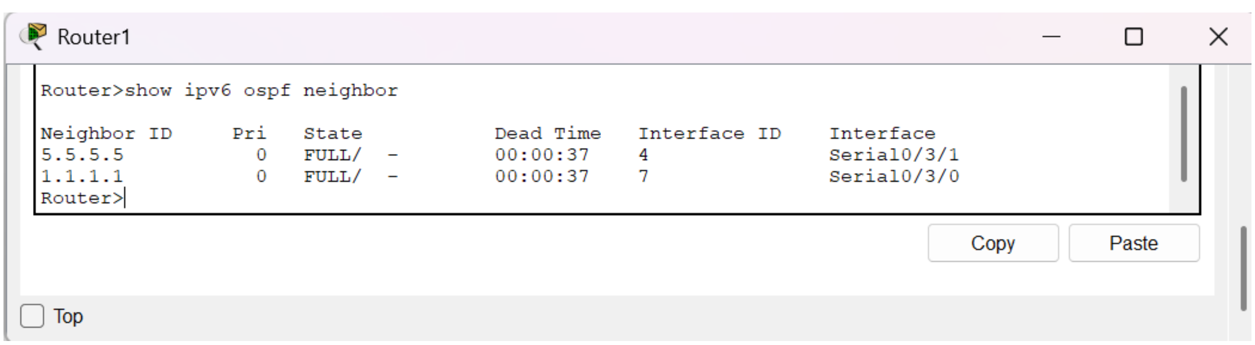
OSPFv3 Peer Verification:

Verify OSPFv3 peer relationships using the show ipv6 ospf neighbor command.



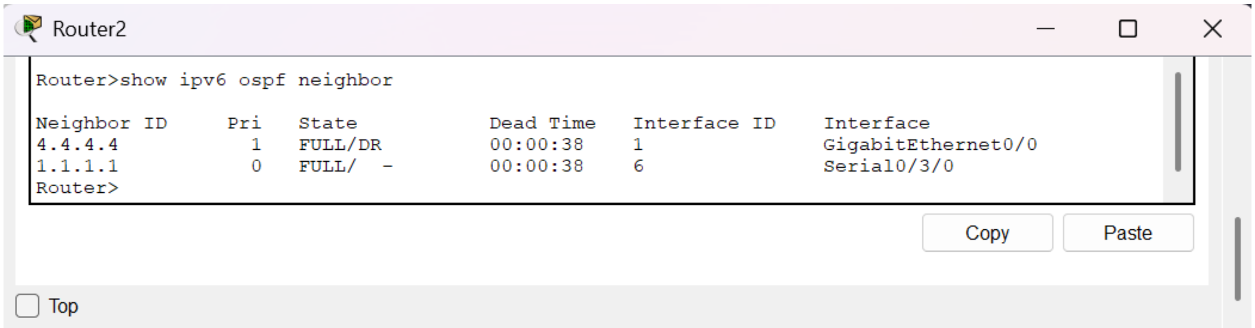

```
Router>show ipv6 ospf neighbor

Neighbor ID     Pri   State           Dead Time   Interface ID    Interface
3.3.3.3           0   FULL/  -        00:00:31    4               Serial0/3/0
2.2.2.2           0   FULL/  -        00:00:34    6               Serial0/3/1
Router>
```



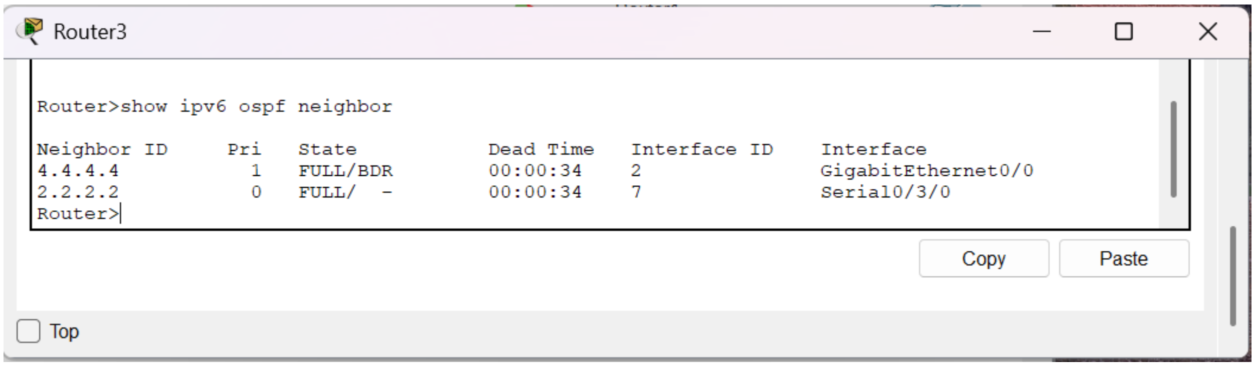

```
Router>show ipv6 ospf neighbor

Neighbor ID     Pri   State           Dead Time   Interface ID    Interface
5.5.5.5           0   FULL/  -        00:00:37    4               Serial0/3/1
1.1.1.1           0   FULL/  -        00:00:37    7               Serial0/3/0
Router>
```




```
Router>show ipv6 ospf neighbor

Neighbor ID     Pri   State           Dead Time   Interface ID    Interface
4.4.4.4           1   FULL/DR         00:00:38    1               GigabitEthernet0/0
1.1.1.1           0   FULL/  -        00:00:38    6               Serial0/3/0
Router>
```




```
Router>show ipv6 ospf neighbor

Neighbor ID     Pri   State           Dead Time   Interface ID    Interface
4.4.4.4           1   FULL/BDR        00:00:34    2               GigabitEthernet0/0
2.2.2.2           0   FULL/  -        00:00:34    7               Serial0/3/0
Router>
```

Ping Test Results:

Conduct ping tests between LANs to ensure end-to-end connectivity. Finally, test connectivity across the network using the ping command:

- LAN 1 to LAN 2: ping 3001:b::1
- LAN 1 to LAN 3: ping 3001:c::2
- LAN 2 to LAN 3: ping 3001:c::1

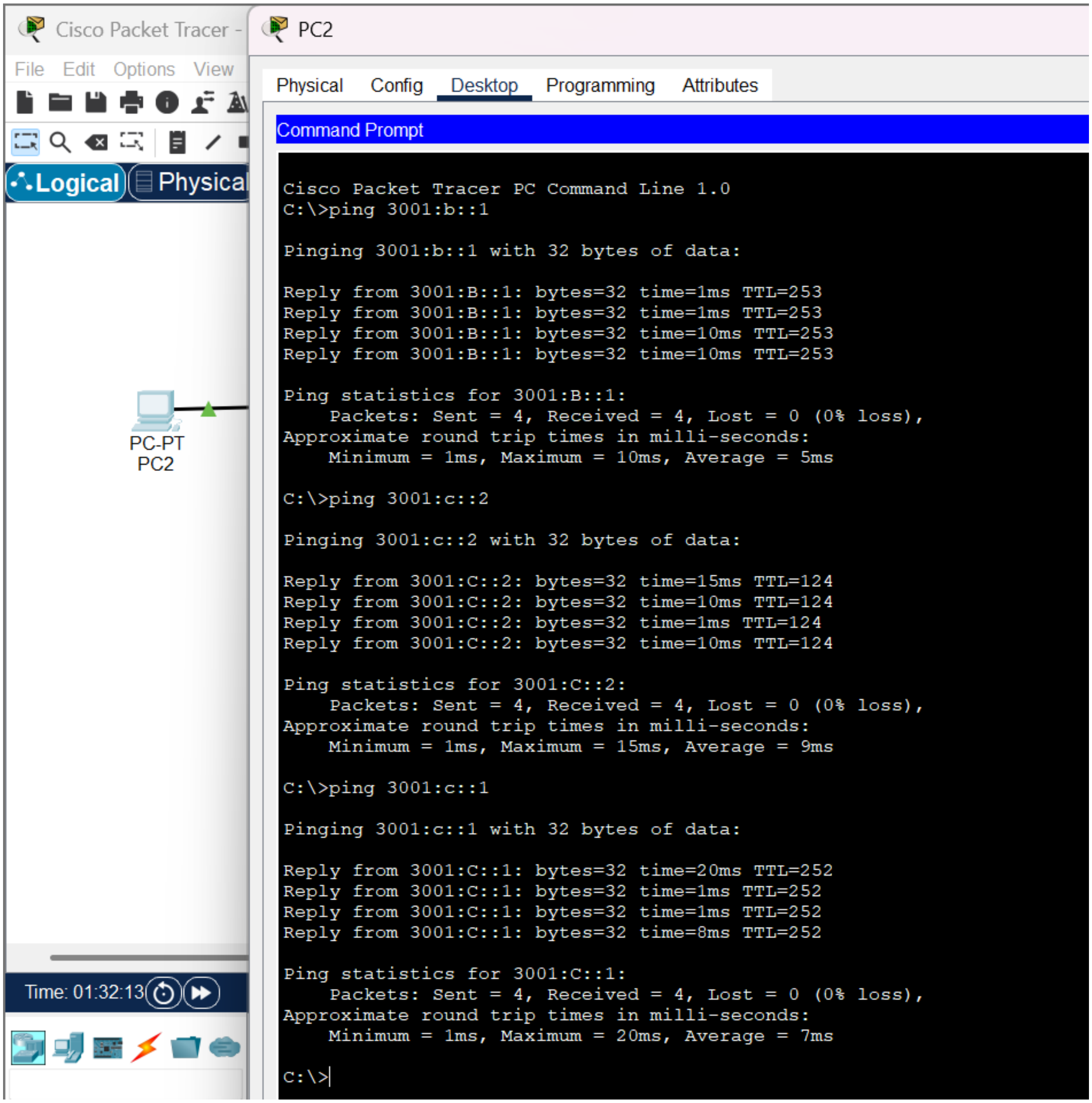


Figure 32: PC2->Desktop->Command Prompt-> ping commands

IPv6 RIP vs. OSPFv3: Conclusion

In the realm of dynamic routing protocols for IPv6, both Routing Information Protocol next-generation (RIPng) and Open Shortest Path First version 3 (OSPFv3) play pivotal roles. In conclusion, the choice between IPv6 RIPng and OSPFv3 depends on the specific requirements and characteristics of the network. For smaller, less complex networks where simplicity is paramount, RIPng may suffice. However, in larger and more intricate networks that demand scalability, faster convergence times, and support for multiple address families, OSPFv3 emerges as a more robust and versatile solution. Ultimately, the selection should align with the network's size, complexity, and future scalability needs.